Local network growth: How simple rules drive network complexity

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To the ICTP, where it all started...

and Nodes & Links, for a renaissance.

Preface

In the year 2000 I was at the International Centre for Theoretical Physics (ICTP), located in Trieste, Italy. I had left my home country, Cuba, and gone there, with many dreams and much nostalgia. At the ICTP I indulged myself in the library, the scientific meetings, the computer cluster, and new friends. Then, on May 22–25, 2000, I attended the Research Workshop on Graph Theory and Statistical Physics. There was a presentation by an obscure Hungarian physicist by the name of Albert-László Barabási: "Emergence of scaling in complex networks." A small conference room and few attendees. Yet, it was eye-opening, a fascinating introduction to the properties of real networks and a hypothesis for their evolution. When I left that room I was charged, full of ideas. Two weeks later I had my first paper on network science: "Knowing a network by walking on it: emergence of scaling" (Vazquez, 2000). And right there, it all started.

Networks are everywhere, and so are the regularities they share. Whether one looks at the Internet, the World Wide Web, a society, a cell, or the scientific literature, the same statistical signatures recur: broad, often scale-free degree distributions; short distances between nodes; high local clustering that thins out in a regular hierarchy as degree grows; and systematic correlations between the degrees of neighboring nodes. That these features appear across systems with nothing else in common is one of the most striking findings of network science. Explaining them has been the field's central occupation for more than two decades.

The dominant explanation has rested on the preferential attachment mechanism, proposed by Réka Albert and Barabási (1999):

new nodes connect to existing ones in proportion to their degree, and a power law follows. It is a powerful idea, but taken literally it is a *global* rule, requiring each newcomer to weigh every node in the network against every other. Real web pages, proteins, and people do no such thing. They act on what is near them — the page a surfer reaches, the friend a friend introduces, the gene that gets copied. Preferential attachment is a macroscopic perception. In the same way, our perception of temperature is a macroscopic consequence of air molecules heating our skin. This book is about the microscopic mechanism behind preferential attachment and beyond: the regularities of real networks *emerge* from local rules. A growth rule that reaches just one step beyond a randomly chosen node, to one of its neighbors, already attaches preferentially by degree — and the same locality that produces preferential attachment produces the clustering hierarchy, the degree correlations, the community structure, and the multiplicity of shortest paths, all as a package. Coincidentally, I consolidated these ideas while working as a postdoctoral research associate in Barabási's group, then at the University of Notre Dame.

The chapters develop this thesis one mechanism at a time. After an introduction that fixes the measures and states the argument, I treat growth by search (citation and web graphs), by triadic closure (social networks), and by duplication (protein interaction networks) — the three local rules of my 2003 paper — and then carry the same principle into more recent work: the duplication–split structure of project schedules, the emergence of communities from local rules, and the boosting of shortest path multiplicity. Each chapter pairs a concrete local rule with a class of real networks it illuminates, and each ends where it began, with the conclusion that locality is the common cause.

The book is intended for anyone curious about how complex systems organize themselves — the interested general reader as much as the specialist. Readers drawn to the *general* principle will find it running through every chapter. Mathematicians, computer scientists, and physicists, in particular, will recognize a single idea applied again and again: that a handful of simple local rules, with no global coordination, is enough to account for the statistics that recur across otherwise unrelated networks; the closing chapters on the emergence of communities and on the redundancy of shortest

paths push that idea furthest. Readers drawn instead to a *particular* system will find it given a chapter of its own. Social scientists will recognize their networks of acquaintance and collaboration in the friends-of-friends mechanism of Chapter 3; biologists will find the evolution of protein interaction networks through gene duplication in Chapter 4; project managers and engineers will find the anatomy of project schedules — and the surprising irrelevance of the critical path — in the activity-specialization model of Chapter 5; and those who study the web, citation, or the spread of information online will find the search-and-connect rule of Chapter 2. Graduate students entering the field will find the models developed in full. Whatever the entry point, the core ideas are told in plain language, and the more technical calculations are set in boxes that can be skipped without losing the thread. I have tried to keep the analysis self-contained, favoring mean-field arguments that make the mechanisms transparent over exhaustive rigor, and to let the empirical data set the agenda throughout. Readers will find the degree denoted k, the elements of a graph called nodes, and their relations called links, following the now-standard usage of network science.

Much of this material grew out of collaborations and conversations over many years, with special mentions to Alessandro Vespignani, Yamir Moreno, Romualdo Pastor-Satorras, Amos Maritan, Alessandro Flammini, Albert-László Barabási, Sara Nadiv Soffer and Christos Ellinas. The original research papers on which the chapters draw are cited at the appropriate points and collected in the bibliography. Parts of the prose, and several of the figures, were prepared and edited with the help of Anthropic's Claude (Opus 4.8). I am grateful to the colleagues who shaped these ideas, and to the reader who, I hope, will come away persuaded that complex networks need not be designed — that they can grow themselves, one local decision at a time.

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Chapter 1

Introduction

1.1 Networks everywhere

Look closely at almost any complex system and you will find, hiding inside it, a network. A network, in the sense used throughout this book, is nothing more exotic than a collection of things together with a record of which pairs of them are connected. The "things" are drawn as dots and called *nodes*. A connection between two of them is drawn as a line and called a *link*. That is the whole vocabulary needed to begin: a network is dots and lines, nodes and links.

What is remarkable is how many different systems fit this simple picture. The Internet is a network whose nodes are routers and whose links are the cables between them. The World Wide Web is a network of pages connected by the hyperlinks you click. A society is a network of people connected by acquaintance, friendship, collaboration, or contact. A living cell contains a network of proteins connected when two of them physically interact, and another of chemicals connected by the reactions that turn one into another. A project is decomposed into discrete activities and the logical dependencies between them. The scientific literature is a network of papers connected by citations, and of authors connected when they write a paper together (Newman, 2001, 2003; Albert and Barabási, 2002; Dorogovtsev and Mendes, 2002). These systems share almost nothing on the surface — one is silicon and glass, another is flesh

and acquaintance, another is ink and ideas — and yet, viewed as networks, they turn out to look astonishingly alike.

That likeness is the subject of this book. Over the past few decades, as researchers gathered maps of real networks ranging from a handful of nodes to billions, the same handful of features kept reappearing, no matter the system. Four of them organize everything that follows, so it is worth meeting them in plain language before we make them precise.

The first is the *small-world* property (Watts and Strogatz, 1998). In a real network you can usually get from any node to any other in only a few hops, even when the network is enormous — the famous "six degrees of separation" between any two people on Earth is one example. At the same time these networks are highly *cliquish*: your friends tend to be friends of one another, far more than chance would predict. Short hops everywhere, plus dense little neighborhoods: that combination is the small world.

The second is the *scale-free* property (Barabási and Albert, 1999; Faloutsos et al., 1999). If you count how many connections each node has, you find that most nodes have very few, while a small number — the *hubs* — have a huge number. Hollywood has a few actors who have worked with almost everyone and a multitude who have appeared in one film; the Web has a few pages everybody links to and billions that are barely linked at all. There is no "typical" number of connections in such a network, which is what the term *scale-free* is meant to capture.

The third, more subtle and the real connective tissue of this book, is that real networks carry systematic *correlations* and a *hierarchy* (Pastor-Satorras et al., 2001; Vazquez et al., 2002; Ravasz and Barabási, 2003; Vazquez, 2002). Whom a node connects to is not random with respect to how connected those neighbors are, and the cliquishness around a node depends in a regular way on how many connections the node has. We will make both of these ideas concrete in a moment.

The fourth, more recent and pitched at a larger scale than the rest, is that real networks divide into *communities*: groups of nodes densely connected among themselves but only loosely tied to the rest. People cluster into social circles, web pages into topics, proteins into functional modules — so that a network often looks less like

one seamless fabric than a patchwork of tight-knit groups stitched loosely together. Detecting these communities, and explaining why they arise, has grown into a research field of its own (Girvan and Newman, 2002; Fortunato, 2010; Palla et al., 2005).

The central question of the book is not *whether* these patterns exist — the evidence is by now overwhelming — but *where they come from*. The answer we will develop, over and over in different settings, is that they need not be designed in. They appear all by themselves in networks that grow according to *local rules*: rules in which each newly arriving node decides where to attach using only what is right in front of it — a node it has already reached, and that node's immediate neighbors. No node ever needs to survey the whole network. That a global pattern can emerge from purely local decisions is, in one sentence, the thesis of this book.

1.2 Some basic measures

To talk about networks precisely we need a few simple measures. None of them is harder than counting; we introduce them gently here and will lean on them in every chapter. Throughout, we picture a network with n nodes. When it helps to be formal we use the *adjacency matrix* a_{ij}, a bookkeeping device that is simply 1 if nodes i and j are connected and 0 if they are not — a giant table of yes/no answers to "are these two linked?"

Degree and degree distribution. The *degree* of a node is its number of links — how many neighbors it has. We write it k, so a node with $k = 3$ has three connections. (In formulas, the degree of node i is $k_i = \sum_j a_{ij}$, which just says: add up the yes-answers in row i of the table.) The *degree distribution* $p(k)$ is the recipe for how degrees are spread across the network: it is the fraction of nodes that have exactly k links, equivalently the chance that a node picked at random has degree k.

A network is called *scale-free* when this distribution follows a *power law*,

$$p(k) \sim k^{-\gamma}, \tag{1.1}$$

where the symbol $\sim$ means "behaves like" for large k, and γ (gamma)

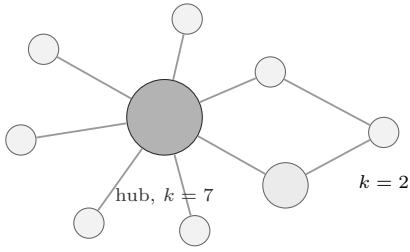


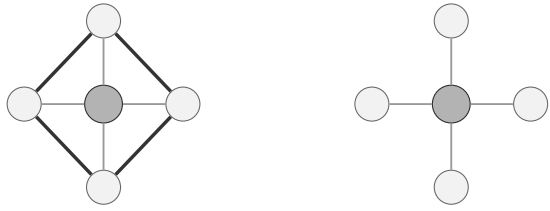
Figure 1.1: **Degree.** A node's degree is simply its number of links. Drawn here with node size growing with degree: a few hubs (such as the central one, $k = 7$) carry many links, while most nodes have only one or two. That imbalance — many small nodes, a few giant hubs — is the hallmark of a scale-free network.

is a number, the *exponent*, typically between 2 and 3 for real networks. A power law is just a rule that says "ten times bigger is some fixed number of times rarer." Unlike the bell-shaped curves of everyday statistics — human heights, say, which cluster tightly around an average — a power law has a long, heavy tail: extremely well-connected hubs are rare but nowhere near as rare as a bell curve would allow. This is why scale-free networks are hub-dominated, and why those hubs end up running much of the show (Figure 1.1).

Clustering coefficient. The *clustering coefficient* measures cliquishness: of all the pairs of your neighbors, what fraction are themselves connected? If you have four friends and four of the possible six friendships among them actually exist, your clustering coefficient is $4/6 \approx 0.67$. Formally, if node i has k_i neighbors and e_i of the possible links among those neighbors are present, then

$$c_i = \frac{2e_i}{k_i(k_i - 1)}, \tag{1.2}$$

since $k_i(k_i - 1)/2$ is the number of pairs that *could* be linked. A clustering coefficient near 1 means a tight clique; near 0 means your contacts are strangers to each other. Averaged over all nodes we write $\langle c \rangle$, where the angle brackets $\langle \cdot \rangle$ are standard shorthand for "average value." The striking empirical fact (Watts and Strogatz,



(a) high: $c \approx 0.67$ (b) low: $c = 0$

Figure 1.2: **Clustering coefficient.** The clustering of the central node (gray) is the fraction of its neighbor pairs that are themselves linked. (a) Four of the six neighbor pairs are linked (the bold links), so $c \approx 0.67$ — a cliquish neighborhood. (b) No neighbors are linked to each other, so $c = 0$.

1998) is that $\langle c \rangle$ in real networks is far larger than you would get by wiring the same nodes up at random — the quantitative fingerprint of those dense local neighborhoods (Figure 1.2).

Average distance. The *distance* between two nodes is the number of links on the shortest route between them — the fewest hops to get from one to the other. Averaged over all pairs of nodes, this distance grows very slowly (logarithmically, or even more slowly) as the network gets bigger. Slow growth of distance, together with high clustering, is precisely the small-world effect named above (Figure 1.3).

Degree correlations. Now to the more subtle pattern. Imagine following a link from a node out to one of its neighbors and asking: how connected is the neighbor I land on? If the network had no *degree correlations*, the answer would not depend on where I started. But real networks are correlated: in some, high-degree hubs tend to sit next to low-degree nodes; in others, hubs cluster together with other hubs. The cleanest way to see this is to record, for nodes of each degree k, the *average degree of their neighbors*, written $\langle k_{nn} \rangle(k)$ (the subscript "nn" stands for nearest-neighbors) (Pastor-Satorras

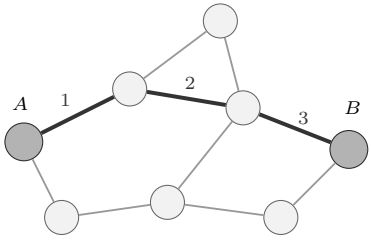


Figure 1.3: **Distance.** The distance between two nodes is the number of links on the shortest route between them. The shortest route from A to B here is three hops (in bold); the lower route is longer. In real networks these shortest distances stay small even as the network grows — the small-world effect.

et al., 2001):

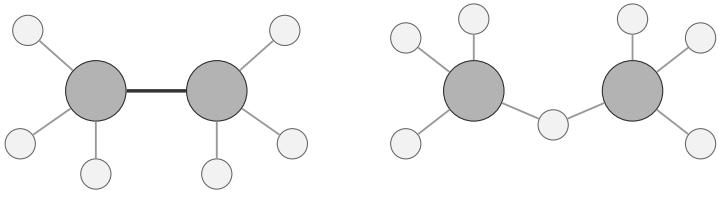
$$\langle k_{nn} \rangle(k) = \sum_{k'} k' \, p(k' \mid k). \tag{1.3}$$

Here $p(k' \mid k)$ is read "the probability that a link from a degree-k node lands on a degree-k' node," and the sum simply averages the landing degree k' weighted by how likely each is. Do not worry if the formula looks abstract; the next box pins down the one fact we need from it, and you can take that fact on faith and skip the box if you prefer.

It is worth knowing what this quantity would be if there were *no* correlations at all, because that is the yardstick against which real networks are judged.

The mathematics behind it: the no-correlation baseline for neighbor degree *(this box can be skipped)*

If degrees are uncorrelated, the only reason a link is more likely to land on a particular node is that the node has more links to land on. The chance of arriving at a node of degree k' is then proportional to k' times how common such nodes are, $k'p(k')$. Averaging the landing degree k' with this weight (and dividing by the total weight



(a) assortative (b) disassortative

Figure 1.4: **Degree correlations.** (a) In an *assortative* network, hubs tend to link to other hubs ("birds of a feather"). (b) In a *disassortative* one, hubs avoid each other and link instead to low-degree nodes ("opposites attract"). Social networks lean assortative; technological and biological ones, disassortative.

to normalize) gives

$$\langle k_{nn} \rangle_{\text{unc}} = \frac{\sum_{k'} k' \cdot k' p(k')}{\sum_{k'} k' p(k')} = \frac{\langle k^2 \rangle}{\langle k \rangle}, \tag{1.4}$$

where $\langle k \rangle$ is the average degree and $\langle k^2 \rangle$ the average of the squared degree.

The result, Eq. (1.4), is the key takeaway: in a network with no degree correlations, the average neighbor degree is the same flat number for every node, regardless of that node's own degree. So when we plot $\langle k_{nn} \rangle(k)$ against k for a real network and the curve is *not* flat, the tilt tells us the kind of correlation present. If the curve slopes *down* — well-connected nodes have poorly-connected neighbors — the network is called *disassortative*; this is typical of technological and biological networks. If it slopes *up* — the well-connected keep company with the well-connected — the network is *assortative*, as social networks tend to be (Newman, 2002). Assortative and disassortative are simply the "birds of a feather" and "opposites attract" of network connectivity (Figure 1.4).

Clustering hierarchy. Finally, we can combine the last two ideas by asking how cliquishness depends on degree. Sorting nodes by

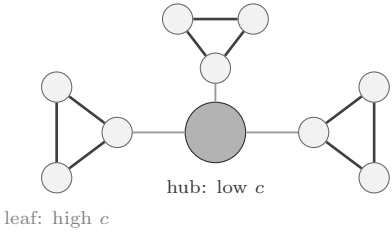


Figure 1.5: **Clustering hierarchy.** Low-degree nodes sit inside tight clusters where everyone is connected (high clustering, the dark triangles), while a hub bridges many such clusters. The hub's own neighbors — one drawn from each cluster — do not know each other, so the hub itself has low clustering. Cliquishness therefore falls as degree rises.

their degree k and averaging the clustering coefficient within each group gives $\langle c \rangle(k)$. In a wide range of real networks this quantity falls off as a power law,

$$\langle c \rangle(k) \sim k^{-1}, \tag{1.5}$$

meaning the more connections a node has, the *less* cliquish its neighborhood. This is the *clustering hierarchy*, and it has an intuitive reading (Ravasz et al., 2002; Ravasz and Barabási, 2003): ordinary low-degree nodes sit inside tight little communities where everyone knows everyone, while the big hubs act as bridges between many such communities, so their own neighbors — drawn from all over — mostly do not know each other. A small-town local has a dense circle of mutual friends; a globe-trotting connector knows thousands of people who have never met (Figure 1.5).

That reading is appealing, and it is the one the literature has largely adopted. It is also, in good part, wrong — and the reason is worth a few pages, because it is the first example in this book of a global pattern dissolving under inspection (Soffer and Vazquez, 2005).

Look again at the denominator of Eq. (1.2). We divided the triangles e_i around a node by $k_i(k_i - 1)/2$, the number of *pairs* of its neighbors, on the grounds that this is how many links those

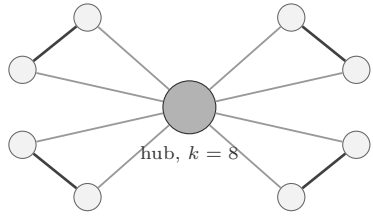


Figure 1.6: **Why the standard clustering coefficient misleads.** The hub's neighbors all have degree two, and each spends its spare link on another neighbor of the hub. They are therefore *as interconnected as their degrees allow* — no rewiring could add a single link among them. Yet the standard coefficient, Eq. (1.2), divides by all $\binom{8}{2} = 28$ neighbor pairs and reports $c = 4/28 \approx 0.14$, falling to zero as the hub grows. The pairs it counts are pairs the degrees forbid.

neighbors could possibly have among themselves. But could they? A neighbor j has only k_j links in total, and one of them is already spent reaching i. So j can reach at most $k_j - 1$ of i's other neighbors — and if k_j is small, that is nearly none of them. The denominator counts pairs that the network's own degrees forbid.

The consequence is not subtle. Consider a hub of degree k joined to k nodes of degree two, each of which spends its second link on one other neighbor of the hub (Figure 1.6). Every neighbor of the hub is now connected to another neighbor: they are as interconnected as their degrees permit, and no rewiring could make them more so. There are $e_i = k/2$ links among them, so the hub's clustering coefficient is

$$c = \frac{2\,(k/2)}{k(k-1)} = \frac{1}{k-1}, \tag{1.6}$$

which tends to zero as the hub grows. By the standard measure this maximally-clustered neighborhood is reported as having no clustering at all. A large-degree node attached to small-degree nodes will always be assigned a small c, however its neighbors are wired — and in a disassortative network, that describes every hub. The measure and the correlations are entangled.

The repair is to divide not by the pairs one can *draw* but by the links the neighbors can *afford*.

The mathematics behind it: clustering without the degree-correlation bias *(this box can be skipped)*

Let ω_i be the largest number of links that could exist among the k_i neighbors of node i, given those neighbors' degrees. A first bound follows from the argument above: neighbor j can spend at most $\min(k_i, k_j) - 1$ links on i's other neighbors — at most $k_j - 1$ because that is all it has left, and at most $k_i - 1$ because that is all the targets there are. Summing over neighbors and halving, since each link is counted from both ends,

$$\omega_i \;\le\; \Omega_i = \left\lfloor \frac{1}{2} \sum_{j \in \mathcal{N}(i)} \big[\min(k_i, k_j) - 1\big] \right\rfloor \;\le\; \binom{k_i}{2}, \qquad (1.7)$$

where $\mathcal{N}(i)$ denotes the neighbors of i. The last inequality is the whole point: Ω_i can be far below $\binom{k_i}{2}$, and it is the standard coefficient's insistence on the larger number that manufactures the hierarchy. Ω_i is still only a bound, because it does not check that the links can actually be arranged; ω_i itself is obtained by trying. Take the list $\min(k_i, k_j) - 1$ over all neighbors, sorted in decreasing order; spend the largest entry's budget on as many of the next-largest entries as it can reach, decrementing each; discard it and any entry now at zero; re-sort and repeat until the list is empty. The number of links drawn is ω_i. The corrected clustering coefficient is then simply

$$\tilde{c}_i = \frac{e_i}{\omega_i}, \qquad (1.8)$$

the fraction of the links the neighbors could afford that they actually made. Three properties follow. It is undefined when $\omega_i = 0$ — a hub of leaves has no clustering to speak of, rather than zero clustering, which is the honest verdict. It never falls below the old one, $\tilde{c}_i \ge c_i$. And when every neighbor is at least as well connected as the node itself, the two coincide, $\tilde{c}_i = c_i$, so nothing is changed where nothing was wrong.

Measure $\tilde{c}$ on real networks and the hierarchy largely evaporates (Figure 1.7, panels b and c, which share a vertical scale so that the

change can be read off directly). For the Internet, $\langle c\rangle(k)$ falls by more than a decade across four decades of degree; $\langle \tilde{c}\rangle(k)$ is *flat* — it varies by a factor of 1.5 over the whole range, and its fitted slope is $+0.02$ where the uncorrected slope was -0.76. The same happens, less completely, for the others: every slope moves sharply toward zero. What is left, when the bias is stripped out, is a clustering coefficient that is either roughly constant or decays only logarithmically with degree (Soffer and Vazquez, 2005) — and a logarithm, over the two or three decades of degree a real network offers, is very nearly a constant.

So Eq. (1.5) should be read with care. It is a robust *measurement*: plot $\langle c\rangle(k)$ for almost any real network and you will see it. It is much weaker as *evidence* for a designed hierarchy of modules, because a large part of what it records is the degree correlations of the network reflected back through a biased denominator. The clustering hierarchy and the degree correlations are not two independent regularities; they are, in good part, two views of one.

None of which makes the clustering itself an illusion. Two facts survive the correction and matter for everything that follows. First, the corrected clustering is *high* — $\tilde{c} \approx 0.39$ for the Internet, 0.80 for co-authorship — so real networks really are cliquish; it is the *trend* with degree, not the cliquishness, that the bias manufactured. Second, and more pointedly, that cliquishness is more than the degrees and their correlations can account for: shuffle the links while holding every degree fixed and the clustering drops, as Table 1.1 will show. The Internet is more clustered than its own degree sequence and degree correlations can explain. Something built those triangles, and the degrees do not know what.

The lesson is to read these measures together rather than in isolation — a theme that recurs throughout the book, where a single local rule will be seen to set the clustering hierarchy and the degree correlations *at once*, as facets of one mechanism. When later chapters derive $\langle c\rangle(k) \sim k^{-1}$ from a local rule, understand the claim precisely: the rule reproduces what the standard measure reports of real networks, biases and all, which is the right target — but the deeper quantity it is matching is the pair of degree correlations and genuine cliquishness that lie underneath.

1.3 From global to local rules

With the measures in hand we can state the puzzle this book exists to solve. The modern story of network growth opens with the model of Barabási and Albert (1999), usually abbreviated BA after its authors. It rests on two ideas. The first is *growth*: networks are not built all at once but accumulate nodes over time, the way the Web gains pages or the literature gains papers. The second is *preferential attachment*: a new node prefers to link to nodes that are already well connected, with a probability proportional to their degree — the "rich get richer" of networks. From these two ingredients a power-law degree distribution follows reliably, and preferential attachment has rightly become the textbook explanation for why so many networks are scale-free.

Two difficulties, though, drive everything that follows. The first is about *mechanism*. Preferential attachment, read literally, is a *global* rule: to attach in proportion to degree, a newcomer would have to know the degree of every node in the entire network and weigh them all against each other. That is implausible for the systems we care about. A scientist writing a new paper has read a few dozen others, not the millions in existence; a new gene is a copy of one existing gene, ignorant of the rest of the genome; a newcomer to a town meets a handful of people, not the whole population. Real nodes act *locally*. So if preferential attachment is not literally true, why do so many networks behave as if it were?

The second difficulty is that the BA model, for all its success with the degree distribution, does not by itself reproduce the other patterns we just met. Its clustering fades away as the network grows, it shows none of the clustering hierarchy of Eq. (1.5), and its degree correlations are weak (Albert and Barabási, 2002; Vazquez, 2003). Something is missing.

The resolution pursued throughout this book is to throw out the global rule and replace it with local ones, then watch what *emerges*. By a *local rule* we mean a way of growing the network in which a newcomer reaches some node — chosen at random, or arrived at by surfing along links, or copied from an existing node — and then connects to that node and perhaps to a few of its immediate neighbors, using nothing but that local information. The recurring

and rather beautiful discovery, established for several such rules (Vazquez, 2003), is that

> *an effective "rich get richer" behavior arises all on its own from growth by local rules.*

The reason, which we will meet many times, is almost a slogan: a well-connected node is, by definition, a node that many local explorations will bump into, because it has many links through which it can be reached. The preference for hubs need not be legislated; it is a free by-product of poking around the network locally. And because the very same local moves also create triangles (your friend introduces you to their friend) and link nodes of unlike degree, the same rules that produce the power law deliver, in one package, the clustering hierarchy of Eq. (1.5) and the degree correlations of Eq. (1.3). Three patterns that look independent turn out to share a single root.

1.4 What the data demand

Any explanation has to answer to measurements, so it is worth recording up front what those measurements actually look like. Rather than take this on trust, let us measure four real networks and see. All four come from the Netzschleuder repository (Peixoto, 2020), a public catalogue of several hundred network datasets, read here through the `graph-tool` library (Peixoto, 2014); the code that fetches and measures them accompanies this book. They are: the *Internet* at the level of the service providers, the "autonomous systems," that own its routers; the *Gnutella* peer-to-peer file-sharing overlay; the protein interaction network of *yeast*; and the *co-authorship* network of the `cond-mat` preprint archive, in which two physicists are linked if they have written a paper together. Three technological or biological, one social — which turns out to be exactly the division that matters.

Start with the small world, since it needs only two numbers per network (Table 1.1). Every one of these networks is small: the average distance between two nodes is between four and seven hops, in networks of up to sixty thousand nodes. And every one is far more cliquish than chance. Wiring the same number of nodes at random with the same number of links — the comparison Watts and Strogatz made — would give the clustering in the column headed

Table 1.1: **Four real networks.** Measured on the largest connected component, undirected. $\langle \tilde{c} \rangle$ is the clustering corrected for what the neighbors' degrees allow, Eq. (1.8); it never falls below $\langle c \rangle$. $\langle c \rangle_{\mathrm{ER}}$ is the clustering of a random network with the same n and $\langle k \rangle$; $\langle c \rangle_{\mathrm{rew}}$ that of the same network with its links shuffled at random but every degree held fixed. $\langle d \rangle$ is the average distance and r the assortativity: negative for the first three, positive for the last.

Network	n	$\langle k \rangle$	$\langle c \rangle$	$\langle \tilde{c} \rangle$	$\langle c \rangle_{\mathrm{ER}}$	$\langle c \rangle_{\mathrm{rew}}$	$\langle d \rangle$	r
Internet (AS)	22 963	4.2	0.230	0.386	0.0002	0.123	3.9	−0.052
Gnutella	62 561	4.7	0.005	0.012	0.0001	0.0002	5.9	−0.036
Yeast proteins	1 458	2.7	0.071	0.215	0.0018	0.0050	6.9	−0.034
cond-mat	36 458	9.4	0.657	0.800	0.0003	0.0018	5.5	+0.055

$\langle c \rangle_{\mathrm{ER}}$, which is between forty and two and a half thousand times smaller than what is observed. The comparison is almost unfair, though, because it also throws away the hubs, and hubs alone create some clustering. The last of the clustering columns is therefore the honest one: it shuffles the links at random while holding every node's degree fixed, and measures what survives. Even against that stricter null the real networks are the more cliquish, by factors of two to three hundred. The cliquishness is real, and it is not merely a shadow of the degree distribution.

Now the three measures that organize the rest of the book (Figure 1.7). The degree distributions are broad and hub-dominated: the Internet runs from nodes with a single link to one with more than two thousand, four decades of degree in a network four decades in size, and the others do the same over a shorter range. Gnutella is the partial exception worth naming — its protocol caps how many peers a client will hold, so its tail is cut short and it is the least scale-free of the four. Real data are like that; the power law of Eq. (1.1) is an idealization that some systems wear better than others.

The clustering hierarchy of Eq. (1.5) is there in all four: $\langle c \rangle(k)$ falls with degree in every case, and the dashed k^{-1} guide shows that the decay is of about the advertised steepness — the measured slopes run from -0.6 to -1.3. Small nodes sit in tight neighborhoods; hubs bridge between them.

And then the two families part company, on the last measure and only on the last measure. For the Internet, for Gnutella and for the

yeast proteins, the average neighbor degree *falls* with degree: these networks are *disassortative*, their hubs presiding over peripheries of poorly-connected nodes rather than huddling together. This is the signature of technological and biological networks (Pastor-Satorras et al., 2001; Maslov and Sneppen, 2002; Jeong et al., 2001). For the co-authorship network it *rises*: prolific, well-connected authors tend to write with one another, and the network is *assortative* (Newman, 2002). The single number r in Table 1.1 says the same thing more briefly — negative for the first three, positive for the fourth — and the sign of that number is the cleanest division in the data.

The contrast between the two families is instructive, because it shows that whatever mechanism we propose must be able to produce *either* sign of correlation, depending on the microscopic move — and, as the chapters will show, local rules can do exactly that. It also tells us the two families are not two kinds of network but one kind seen through one dial: everything else in Figure 1.7 they share.

These regularities are robust, they hold across many orders of magnitude in network size, and for years their shared origin was a mystery. Supplying that origin, one mechanism and one class of networks at a time, is the work of this book.

1.5 Plan of the book

The chapters proceed from mechanism to application, each built around one concrete local rule and a family of real networks it illuminates. The reader need not take any chapter on faith: in each, a simple local rule is stated, its consequences are worked out (with the heavier algebra confined to boxes that can be skipped), and the predictions are held up against real data.

- **Search and connect.** A newcomer learns of existing nodes by exploring the network — following references, surfing hyperlinks, performing a random walk — and connects to what it finds. This is the natural model for searchable, directed networks such as the citation network and the World Wide Web, and it is where we show in detail how the simple act of surfing manufactures an effective "rich get richer."

- **Friends of friends.** A newcomer connects to a node and then to some of that node's neighbors — you meet a person, then their friends. This rule for social network growth creates triangles directly, and it generates the assortative correlations and the cliquishness hierarchy of social networks.

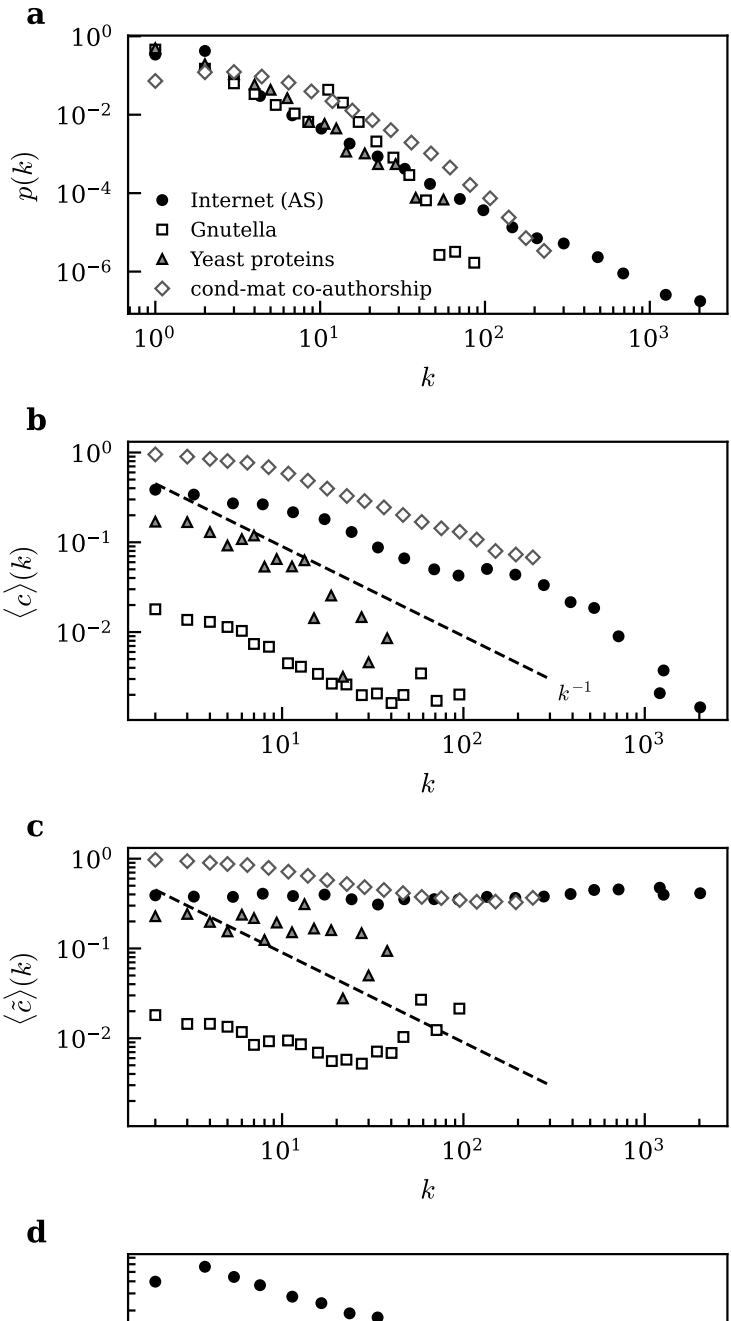
- **Gene duplication.** Biological networks grow by copying: a node is duplicated together with its connections, after which links are gained and lost. Duplication is an intrinsically local move, and it explains the disassortative, hierarchical structure of protein interaction networks.

- **Activity specialization.** The schedule of a large project is a network of activities grown by a local refinement rule: each activity is either duplicated into parallel copies or split into specialized sequential parts. The same copying mechanism gives power-law degrees, while making the celebrated "critical path" vanish as projects grow and relocating delay risk in the network as a whole.

- **Emergence of communities.** Local rules do not only set degrees and correlations; they also organize nodes into communities — tight groups with many links inside and few outside. We examine how this mesoscale structure arises from the same local dynamics, even when every node is otherwise identical.

- **Boosting shortest path multiplicity.** Finally we turn to the redundancy of routes through a network — how many equally short paths join two nodes — and show that local growth multiplies them, tying this redundancy to the very community structure of the previous chapter.

A single thread runs through all of them. In each case the apparently global features of a network — its scale-free degrees, its small-world distances, its clustering hierarchy, its degree correlations, its communities, its redundant paths — turn out to be the collective, emergent consequence of simple rules that no node applies with more than local knowledge. That is the sense in which the title of this book should be read: not networks built by a designer, but networks that grow themselves, one local decision at a time.


a
10^0
10^{-2}
10^{-4}
10^{-6}
$p(k)$
Internet (AS)
Gnutella
Yeast proteins
cond-mat co-authorship
10^0
10^1
10^2
10^3
k
b
10^0
10^{-1}
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$\langle c \rangle(k)$
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10^2
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10^{-2}
$\langle \tilde{c} \rangle(k)$
10^1
10^2
10^3
k
d

Chapter 2

Search and connect

2.1 Finding what to link to

The introduction left us with a puzzle. The "rich get richer" rule explains why so many networks are scale-free, but as a literal recipe it asks the impossible: a newcomer would have to know the degree of every node in the network in order to favor the well-connected ones. In this chapter we solve the puzzle for an important family of networks — the *searchable* ones — by replacing that impossible global rule with the most natural local act a newcomer can perform. It *searches* the network, moving from node to node along the links, and connects to what it happens to find.

The systems to keep in mind are collections of documents that point at one another. In the *citation network*, each node is a published article and each link is a reference from one article to another. In the World Wide Web, each node is a page and each link is a hyperlink. These are *directed* networks: a reference, like a hyperlink, points one way. An article cites an older one, not the reverse; a page links to another, which need not link back. Because direction matters, each node now has two kinds of degree. Its *in-degree*, written k^{in}, counts the links pointing *at* it — how many times a paper has been cited, how many pages link to a given page. Its *out-degree*, written k^{out}, counts the links pointing *out* of it — how many references a paper makes, how many hyperlinks a page contains. It is the

in-degree that interests us, because in real document networks it is the in-degree — the count of incoming attention — that is broadly distributed, with a few hugely-cited classics and a long tail of papers cited once or never.

How does a new document come to point at the old ones? Not by consulting a global popularity ranking. An author writing a new paper finds references by reading a few papers and following *their* reference lists, and then the references of those, and so on (Price, 1965). A web author discovers pages by surfing from link to link and by typing queries into a search engine (Brin and Page, 1998). In every case the information used is strictly local — a document one has reached, and the documents it points to. The claim of this chapter is that this local searching, repeated by every newcomer, adds up to exactly the "rich get richer" behavior that the introduction had to assume by fiat, and that it brings the power-law degrees, the clustering hierarchy, and the degree correlations along with it (Vazquez, 2001, 2003).

2.2 Surfing as a random walk

To turn "surfing" into something we can analyze, we model the surfer as a *random walker* on the network. A random walk is just what it sounds like: start somewhere, and step to a randomly chosen neighbor, then to a neighbor of *that* node, and so on. Our surfer does this with a twist. From whatever page it is on, with probability q_e it follows one of that page's outgoing hyperlinks, picked at random; with the remaining probability $1 - q_e$ it gets bored, stops following links, and jumps to a completely random page elsewhere. The single dial q_e, a number between 0 and 1, sets how patiently the surfer explores along links before restarting from scratch.

Now ask: in the long run, what fraction of its time does the walker spend at each page? Call that fraction the page's *visiting probability*, v_i for node i. A page is visited often if many links point to it and those linking pages are themselves visited often — a self-referential condition that turns out to have a clean form. The remarkable thing, worth flagging before any algebra, is that this visiting probability is *exactly the quantity that Google was built on*: it is the *PageRank* of the page, the measure of importance that ranks search results (Brin

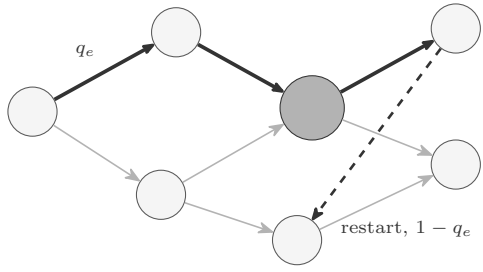


Figure 2.1: **Surfing as a random walk.** From the page it is on, the searcher follows one of the outgoing links with probability q_e (solid arrows) or, with probability $1 - q_e$, jumps to a page chosen at random (dashed arrow). A page reached along many routes — like the highlighted hub — is visited most often; this visiting frequency is the page's PageRank, and it grows in proportion to the number of links that point at the page.

and Page, 1998). So the local rule we are about to write is not a toy — it is the very process real search engines compute and then display to the authors who go on to create new links.

The picture to keep in mind is Figure 2.1. We would like to know how a page's visiting probability depends on its in-degree. Getting there exactly is hard, so we make the standard physicist's move of replacing fluctuating quantities by their averages — a so-called *mean-field approximation*, "mean" as in average. The next box carries this out; the one sentence to carry away is stated right after it, so the box can be skipped.

The mathematics behind it: from surfing to the visiting probability *(this box can be skipped)*

The visiting probability of node i obeys a balance condition: the walker arrives at i either by a random jump (probability $1 - q_e$, landing on any of the n nodes equally) or by following a link into i from some node j that points to it,

$$v_i = \frac{1 - q_e}{n} + q_e \sum_j a_{ij} \frac{v_j}{k_j^{out}}, \tag{2.1}$$

where $a_{ij} = 1$ when j points to i, and the factor $1/k_j^{out}$ accounts for j spreading its attention over its k_j^{out} outgoing links. This is the PageRank equation. To simplify it, replace the sum over the specific nodes pointing to i by an average. A node is reached along an incoming link from a node of out-degree k^{out} with probability $k^{out}p_{k^{out}}/\langle k^{out}\rangle$, and that link is then followed with probability $1/k^{out}$; the two factors of k^{out} cancel, so each incoming link contributes the same average amount $\Theta = \langle v\rangle/\langle k^{out}\rangle$. Since node i has k_i^{in} incoming links,

$$v_i = \frac{1 - q_e}{n} + q_e\,\Theta\,k_i^{in}. \tag{2.2}$$

The payoff is Eq. (2.2), and it is simple to state in words: a page's visiting probability grows in a straight line with its in-degree. Double the number of links pointing at a page and you roughly double how often the surfer lands on it. That straight-line relationship is the seed from which everything in this chapter grows.

2.3 An effective "rich get richer"

Being visited is not the same as being linked to — you read many pages you do not cite. So suppose that each time the surfer visits a page it creates a new incoming link to it only with some probability q_v. We can then ask how fast a page accumulates new links as a function of how many it already has. This rate is the network-growth analog of preferential attachment, and computing it is the crux of the chapter. The derivation is short; its conclusion is in the sentence after the box.

The mathematics behind it: the rate at which a page gains links *(this box can be skipped)*

Let ν_s surfers be active and ν_a new pages be added per unit time. The node count n and link count e grow as

$$\frac{\partial n}{\partial t} = \nu_a, \qquad \frac{\partial e}{\partial t} = \nu_s\,q_v\,\langle v\rangle\,n, \tag{2.3}$$

because each of the $\langle v\rangle n$ visits in a walk makes a link with probability q_v. Integrating fixes the average degree and hence $\Theta = \nu_a/(q_v\nu_s n)$.

The rate at which a page of in-degree k^{in} gains its next link is $A(k^{in}) = q_v\, v(k^{in})$; substituting Eq. (2.2) and the value of Θ,

$$A(k^{in}) = \frac{1}{n}\left[\, q_v(1-q_e) + q_e\,\frac{\nu_a}{\nu_s}\,k^{in}\right]. \tag{2.4}$$

Equation (2.4) is the central result, and it deserves to be read slowly. The rate at which a page attracts new links has two parts: a constant piece, the same for every page, which gives even a brand-new page a chance of being noticed; and a piece *proportional to its current in-degree*. That second piece is preferential attachment — the rich getting richer — but notice where it came from. The surfer never consulted a global ranking of pages; it only followed links, one at a time. The bias toward well-connected pages appeared on its own, for the simple reason that a page with many incoming links is a page the wandering surfer reaches by many routes. We have *derived* the rule the introduction had to assume. The same logic underlies the "redirection" trick of Krapivsky and Redner (2001), in which a newcomer points either to a random node or to whatever that node points to — a one-step walk that yields the same effective preference.

2.4 The degree distribution

A linear attachment rate is exactly the ingredient needed to produce a power law. Working out the resulting distribution of in-degrees requires solving a balance equation for how many pages sit at each in-degree, which we do in the box. Readers willing to take the outcome on trust can jump to the discussion after it.

The mathematics behind it: the in-degree distribution and its exponent *(this box can be skipped)*

Let $n_{k^{in}}$ be the number of nodes with in-degree k^{in}. It changes as nodes climb from one in-degree to the next by gaining links, with

new nodes injected at in-degree zero:

$$\frac{\partial n_{k^{in}}}{\partial t} = \nu_s\, A_{k^{in}-1}\, n_{k^{in}-1} - \nu_s\, A_{k^{in}}\, n_{k^{in}} + \nu_a\, \delta_{k^{in},0}\,, \tag{2.5}$$

the symbol $\delta_{k^{in},0}$ being 1 only at $k^{in} = 0$. For a network growing at a constant rate the ratio $\nu_s/\nu_a = \alpha$ is fixed and the shape of the distribution settles down, $n_{k^{in}}(t) = N p_{k^{in}}$. Solving for that stationary shape gives (with Γ the gamma function, a smooth generalization of the factorial)

$$p_{k^{in}} = \frac{1}{1+a}\,\frac{\Gamma[a(\gamma-1)+k^{in}]}{\Gamma[a(\gamma-1)]}\,\frac{\Gamma[(1+a)(\gamma-1)+1]}{\Gamma[(1+a)(\gamma-1)+k^{in}+1]}, \tag{2.6}$$

$$\gamma = 1 + \frac{1}{q_e}, \qquad a = \alpha\, q_v\,(1-q_e), \tag{2.7}$$

which for large in-degree is the power law

$$p_{k^{in}} \sim (k^{in})^{-\gamma}. \tag{2.8}$$

Two features of the result, Eq. (2.7), are worth dwelling on. First, the exponent γ is at least 2 — exactly the range seen in real citation and web networks. Second, and more surprising, γ depends *only* on q_e, the surfer's patience for following links. It does not depend on q_v, how often readers actually cite what they read, nor on the growth rate of the network. Whether authors are generous or stingy with citations changes how many links there are, but not the *shape* of the distribution; only the manner of *exploring* the network does that. As the surfer explores ever more deeply ($q_e \to 1$) the exponent drops to its minimum of 2, the most hub-dominated case; as the surfer restarts constantly ($q_e \to 0$) the exponent grows without bound and the heavy tail of hubs disappears.

2.5 Cliquishness and correlations

The same walk that sets the degrees also lays down triangles, so it predicts the clustering hierarchy of the introduction with no extra assumptions. The reason is simple to picture: whenever the surfer steps from a page to one of that page's neighbors and links to both, it has closed a triangle. A short calculation turns this picture into a

formula.

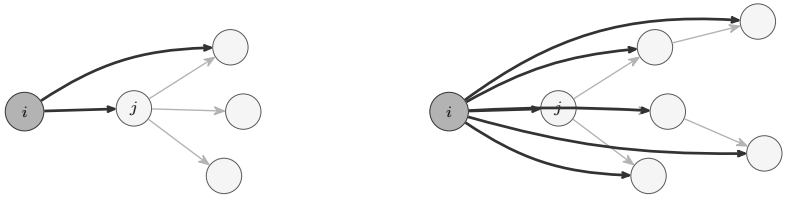
The mathematics behind it: how clustering falls with degree *(this box can be skipped)*

Counting the links e_i among the neighbors of node i created by the walk gives $\partial e_i/\partial t \approx (1+q_e)\,\partial k_i^{in}/\partial t$. Integrating this differential equation, assuming $e=0$ for $k=1$, we obtain $e=(1+q_e)(k-1)$. Inserting this into the definition of the clustering coefficient, $\langle c\rangle_k = 2e(k)/[k(k-1)]$, yields

$$\langle c\rangle_k \approx \frac{2(1+q_e)}{k}. \tag{2.9}$$

The message of Eq. (2.9) is that cliquishness is inversely proportional to degree: a node with twice the degree has roughly half the clustering. This is precisely the clustering hierarchy of Eq. (1.5), seen in the Internet, the Web, and the protein network — and here it falls out for free from growth by searching. The intuition is that a node gains a triangle every time the surfer steps from it to a neighbor, so its triangles grow in step with its degree, while the number of *possible* triangles grows much faster, like the square of the degree; their ratio, the clustering, therefore falls off as one over the degree.

The degree correlations are governed by the same single dial q_e. Numerical simulation of the model shows that when q_e is small the network looks essentially uncorrelated — the average neighbor degree is flat, as in the baseline of Eq. (1.4) — while as q_e grows the average neighbor degree develops the downward slope of a *disassortative* network, hubs sitting amid poorly-connected pages. This matches the real Web, whose measured exponent $\gamma \approx 2.1$ corresponds through Eq. (2.7) to a patient surfer with $q_e > 0.5$, squarely in the disassortative regime. One knob, q_e, thus fixes the degree-distribution exponent, the cliquishness hierarchy, and the sign of the correlations all at once — tying together what the introduction presented as three separate facts.



(a) random-walk model (b) recursive-search model

Figure 2.2: **Two ways to search, when a new node i joins.** (a) In the random-walk model, i links to a randomly chosen node j and then takes a single step to one of j's neighbors, linking to it as well. (b) In the recursive-search model, i follows *all* of j's links, and then their links in turn, linking to every node its widening search reaches. New links are drawn in bold, pre-existing links thin and gray.

2.6 Two ways to search

It helps to see two concrete versions of the searching rule, because they behave differently and bracket the range of possibilities. Figure 2.2 contrasts them.

The random-walk model. The simplest version uses a single surfer that always links what it visits and starts a fresh search from a new node each time it stops. Stated as a recipe, beginning from a couple of connected nodes, repeat:

- *Add.* Create a new node with one link pointing to a randomly chosen existing node.
- *Walk.* Having just linked to a node, then with probability q_e also link to one of that node's neighbors; otherwise go back to the add step.

Here the surfer takes a single step along the network before possibly stopping. This model produces the power-law tail in Eq. (2.8) and the clustering law of Eq. (2.9) (the latter in very good agreement with simulation), while the predicted exponent is a slight overestimate — the price of the mean-field approximation in the box above.

The recursive-search model. A more thorough searcher does not stop after one step but follows *all* the links of each node it reaches, then all the links of those, exploring an ever-widening territory (Vazquez, 2001). Specifically,

- *Add.* Create a new node with one link pointing to a randomly chosen existing node.
- *Walk.* If a link is created to a node in the network then with probability q_e a link is also created to each of its nearest neighbors. When no link is created go to the add rule.

This version has a richer behavior — in fact a genuine *phase transition*, a sudden change of character as the dial q_e is turned, of the kind familiar from water freezing into ice. The next box records the two cases that can be solved exactly and the location of the transition; the meaning is summarized afterward.

The mathematics behind it: the recursive searcher and its phase transition *(this box can be skipped)*

For $q_e = 0$ only the add step operates, every node is equally likely to be linked and $A(k^{in}) = 1/n$ independently of k^{in}. In this case the steady state solution $n_{k^{in}}(n) = n\, p_{k^{in}}$ of Eq. (2.5) is

$$p_{k^{in}} = 2^{-(k^{in}+1)}. \tag{2.10}$$

For $q_e = 1$ the recursion reaches every neighbor, the attachment rate becomes exactly linear, $A(k^{in}) = (1 + k^{in})/n$ — genuine preferential attachment, generated locally — and the steady state solution $n_{k^{in}}(n) = n\, p_{k^{in}}$ of Eq. (2.5) is

$$p_{k^{in}} = \frac{1}{(k^{in}+1)(k^{in}+2)} \sim (k^{in})^{-2}. \tag{2.11}$$

In between, we don't have an analytical solution. Yet, the average out-degree in the steady state is

$$\langle k^{out} \rangle = \frac{1 - \sqrt{1 - 4q_e}}{2q_e}, \tag{2.12}$$

which is finite only for $q_e \leq q_c = 1/4$; above this threshold it diverges.

The exact results bracket the behavior: a fast-decaying exponential when the searcher barely explores, and a clean power law with exponent 2 when it explores exhaustively. The new feature is the threshold q_c in Eq. (2.12). Below it the network has a well-defined average connectivity; above it the average degree runs away and the in-degree distribution locks onto the power-law exponent $\gamma = 2$, no matter the precise value of q_e. Simulations confirm this picture, with the threshold near $q_c \approx 0.5$.

The striking part is that scale-invariance — the power law — is not produced by carefully tuning q_e to one magic value, but holds across a whole range of q_e above the threshold. Physicists have a name for systems that sit automatically at such a critical, scale-free point without fine-tuning: *self-organized criticality*, the textbook example being a sandpile that, however you pour sand on it, keeps adjusting itself to the brink of avalanching. Here the slow addition of new pages and the fast exploration of the existing network play the roles of slow pouring and fast avalanching. Read this way, the ubiquity of scale-free citation and web networks is not a delicate coincidence but the generic, self-organized outcome of building a network by searching it.

2.7 Summary

Searchable networks grow by a rule with nothing global in it: a newcomer explores the existing network along its links and connects to what it finds. Modeling the explorer as a random walker — the very PageRank process that ranks web pages — we found that a page's visiting probability rises in a straight line with its in-degree, which converts the local act of walking into an effective "rich get richer." From that single mechanism follow a power-law in-degree distribution whose exponent $\gamma = 1 + 1/q_e$ is set purely by how the network is explored, a clustering hierarchy in which cliquishness falls as one over the degree, and tunable, generically disassortative degree correlations. Letting the searcher explore exhaustively turns the smooth picture into a self-organized critical transition with a robust exponent of 2. The puzzle of where preferential attachment "comes from," for this family of networks, is thereby dissolved: it comes from walking.

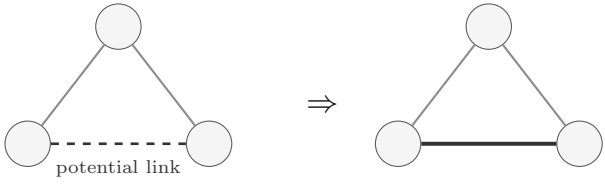
Chapter 3

Friends of friends

3.1 Triadic closure as a growth rule

The previous chapter built networks by exploration. This one builds them by introduction. Think about how you came to know the people you know. Only rarely did you befriend a complete stranger; far more often a friend introduced you to one of *their* friends. This everyday mechanism — a new tie forming between two people who already share a common acquaintance — has a name in sociology: *triadic closure*, because it "closes" a triangle by adding the third side to a path of two. Two people who both know a third are, in a sense, a triangle waiting to happen, and triadic closure is the event that completes it. Sociologists have long seen this as the engine of social structure (Granovetter, 1973). If most new links close triangles, then triangles pile up, and the network acquires the high cliquishness — the large clustering coefficient of the introduction — that sets social networks apart from random ones.

Figure 3.1 shows this elementary step. Notice that it is a thoroughly *local* rule. To introduce two friends you need to know only one node and its neighbors: who you know, and who they know. That is exactly the kind of local information this book is built on, and the question, as always, is what such a rule produces in the aggregate. The answer, worked out below following Vazquez (2003) and reviewed in Vazquez (2002), is that connecting friends-of-friends



(a) friend of a friend (b) triangle closed

Figure 3.1: **Triadic closure.** Two people (bottom) who already share a common friend (top) form a *potential link* — a friend-of-a-friend pair, drawn dashed. Closing the triangle turns that potential link into a real one, the elementary step by which the friends-of-friends rule builds social structure.

generates the whole suite of social-network signatures at once: an effective "rich get richer" and hence a power-law degree distribution, the clustering hierarchy in which cliquishness falls with degree, and — in pointed contrast to the technological and biological networks of the other chapters — *assortative* degree correlations, the tendency of well-connected people to know one another.

Models in this spirit have a history. Davidsen et al. (2002) introduced an acquaintance-network model (the DEB model, after its authors) in which people are repeatedly introduced to friends of friends, producing a small world with high clustering; Holme and Kim (2002) bolted an explicit triangle-forming step onto preferential attachment to give it tunable clustering. What follows organizes these ideas around a single bookkeeping device that makes the mechanism transparent — the *potential link*.

3.2 Potential links

The key idea is that, under triadic closure, a pair of nodes is not just "connected" or "not connected." A pair that already shares a common neighbor is a triangle waiting to close — a candidate in a way that an arbitrary pair of strangers is not. So we sort every pair of nodes into one of three states:

- *disconnected* (s): no link and no common neighbor — two strangers with no mutual acquaintance;
- connected by a *potential link* (p): no link yet, but at least one common neighbor — two friends-of-a-friend, poised to be introduced;
- connected by a *link* (e): already friends.

The *potential link* is the formal stand-in for "friend of a friend." We write k_i for the ordinary degree of node i (its number of friends) and k_i^* for its *potential degree*, the number of friends-of-friends it has — the number of potential links waiting at it.

Triadic closure is then a story about a pair changing state: a potential link (p) becomes a real link (e) when the triangle closes. To track this we need to know the rates at which pairs hop between the three states, and one relationship among those rates is the mechanical heart of the model. It says that potential links are born in proportion to a node's degree — introduce yourself to someone with many friends and you instantly acquire many friends-of-friends. The next box sets up the bookkeeping and draws out its main consequence; the conclusion is restated plainly afterward, so the box may be skipped.

The mathematics behind it: from the three-state bookkeeping to "rich get richer" *(this box can be skipped)*

Let $\nu_{x\to y}$ be the rate (per unit of network size n) at which a pair hops from state x to state y. Ignoring fluctuations, the average degree and potential degree of node i evolve as

$$\begin{aligned} n\,\frac{\partial k_i}{\partial n} &= \nu_{s\to e}\,\hat{k}_i + \nu_{p\to e}\,k_i^* - (\nu_{e\to s} + \nu_{e\to p})\,k_i, \\ n\,\frac{\partial k_i^*}{\partial n} &= \nu_{s\to p}\,\hat{k}_i + \nu_{e\to p}\,k_i - (\nu_{p\to s} + \nu_{p\to e})\,k_i^*, \end{aligned} \tag{3.1}$$

where $\hat{k}_i = n - k_i - k_i^*$ is the number of nodes that are neither friends nor friends-of-friends of i. The link between the two kinds of connection comes from the geometry of triangles: when a new node i attaches to an existing node j, it instantly shares the common neighbor j with *all* of j's neighbors, gaining a potential link to each.

Hence potential links are spawned in proportion to degree,

$$\nu_{s\to p} = \nu_{s\to e}\, k_i, \qquad \nu_{p\to s} = \nu_{e\to s}\, k_i. \tag{3.2}$$

Now make two modeling choices. First, neglect link deletion, $\nu_{e\to s} = 0$ — accurate for, say, a co-authorship network, where having written a paper together is a fact that cannot later become false. Second, make closing a triangle far likelier than linking two strangers,

$$\nu_{s\to e} = \frac{\mu_0}{n^2}, \qquad \nu_{p\to e} = \frac{\mu_1}{n}, \tag{3.3}$$

the stranger-linking rate carrying an extra factor of $1/n$. Substituting these into Eq. (3.1) collapses the system to

$$n\,\frac{\partial k_i}{\partial n} = \mu_0 + \mu_1\, k_i^*, \qquad n\,\frac{\partial k_i^*}{\partial n} = \mu_0\, k_i - \mu_1\, k_i^*. \tag{3.4}$$

The upshot of Eq. (3.4) is that a node's degree grows at a rate proportional to how many friends-of-friends it has — and a node with many friends has, through them, many friends-of-friends. So once again the rich get richer, but this time the preference is manufactured by triadic closure rather than by surfing: a well-connected person is simply involved in more potential introductions. As in the previous chapter, no node ever consulted a global ranking; the bias toward hubs is a free by-product of a purely local act.

3.3 Degree distribution and clustering hierarchy

Because the growth rate is linear in degree, the same reasoning that gave a power law in the last chapter applies here, and it simultaneously fixes the cliquishness. The box carries out both; the two results and their meaning are in the paragraph after it.

The mathematics behind it: the power law and the clustering hierarchy *(this box can be skipped)*

Integrating the linear system (3.4) gives power-law growth of both degrees with the age of the node,

$$k_i(n) = k_0 \left(\frac{n}{n_i}\right)^{\beta}, \qquad k_i^*(n) = k_0^* \left(\frac{n}{n_i}\right)^{\beta}, \tag{3.5}$$

where n_i is the network size when node i arrived and

$$\beta = \frac{\mu_1}{2}\left(-1 + \sqrt{1 + 4\,\frac{\mu_0}{\mu_1}}\right). \tag{3.6}$$

With nodes added at a constant rate, $P(n_i{=}n) = 1/n$, a node has degree above k only when it is old enough for Eq. (3.5) to exceed k; averaging over its birth time gives the cumulative distribution

$$P(k_i > k) = P\left[k_0 \left(\frac{n}{n_i}\right)^{\beta} > k\right] = \int_0^n \frac{dn_i}{n}\,\Theta\left[k_0 \left(\frac{n}{n_i}\right)^{\beta} - k\right] \tag{3.7}$$

with $\Theta[\cdot]$ the step function, 1 when its bracket holds and 0 otherwise. Differentiating it leaves a power-law degree distribution,

$$p_k = \frac{\partial P(k_i > k)}{\partial k} \sim k^{-\gamma}, \qquad \gamma = 1 + \frac{1}{\beta}. \tag{3.8}$$

For the clustering, the dominant way node i gains a triangle is a potential link closing: when the potential link from i to j (sharing neighbor k) becomes real, i gains both the neighbor j and a link j–k among its neighbors. Thus $\partial e_i/\partial n = \nu_{p\to e}\, k_i^* = \mu_1 k_i^*/n$, and integrating with Eq. (3.5),

$$\langle c \rangle_k = \frac{2e(k)}{k(k-1)} \approx \frac{2\mu_1}{k}. \tag{3.9}$$

Two familiar results emerge. First, Eq. (3.8): the degree distribution is again a power law. It is worth knowing exactly where the power law comes from — it comes from the proportionality in Eq. (3.2), the fact that potential links are spawned in proportion to degree. Were they instead spawned at a flat rate, the mechanism would still create triangles but the degree distribution would

decay exponentially, with no fat tail of hubs. Scale-freeness is the fingerprint of that degree-proportional spawning. Second, Eq. (3.9): cliquishness again falls as one over the degree — the clustering hierarchy of the introduction, now arriving by a third route (after search, and before duplication in the next chapter). Here its origin is especially vivid: every triangle a node owns is just a friend-of-a-friend pair that has been introduced.

3.4 Connecting nearest neighbors

The continuum theory is borne out by a minimal simulation, a growing version of the DEB rule (Davidsen et al., 2002). Starting from a single node and no links, repeat:

- with probability $1-u$, *add* a new node and link it to a randomly chosen node j — which by definition creates a potential link between the newcomer and every neighbor of j;
- with probability u, *close* a triangle: pick a potential link at random and turn it into a real link.

The single dial u is the rate of triangle-closing relative to adding newcomers. Feeding the mapping $\mu_0 = 1$, $\mu_1 = u/(1-u)$ into the formulas above gives the exponent in closed form, as the short box records.

The mathematics behind it: the exponent as a function of the closure rate *(this box can be skipped)*

With $\mu_0 = 1$ and $\mu_1 = u/(1-u)$, Eqs. (3.6) and (3.8) give

$$\gamma(u) = 1 + \frac{2(1-u)}{u}\left(-1 + \sqrt{1 + 4\,\frac{1-u}{u}}\right)^{-1}, \tag{3.10}$$

with limiting values

$$\gamma(0) = \infty, \qquad \gamma(1) = 2. \tag{3.11}$$

In words, the exponent is tunable by how eagerly the network closes triangles. When closure is frequent and newcomers rare

($u \to 1$) the exponent sinks to its minimum of 2, the most hub-dominated case; when newcomers arrive so fast that triangles never get a chance to close ($u \to 0$) the exponent blows up and the heavy tail of hubs is lost. Simulations up to a million nodes confirm the power-law degree distribution (the mean-field formula slightly overestimating the exponent, as in the previous chapter), and they confirm the clustering hierarchy, with cliquishness falling as a power of degree.

3.5 Why social networks are assortative

The most telling prediction is about the degree correlations — and here the friends-of-friends rule parts company with every other rule in this book. Measuring the average neighbor degree $\langle k_{nn} \rangle(k)$ in the simulated network, one finds it *rises* with degree: the well-connected are connected to the well-connected. The network is *assortative* (Newman, 2002) — "birds of a feather" — exactly as measured in real collaboration and co-authorship networks, and exactly opposite to the disassortative Internet and protein networks of the introduction.

Why the opposite sign? The reason becomes obvious once seen. Every new link here closes a triangle, joining two nodes that share a common neighbor. But a node that serves as the common neighbor of many pairs is, by definition, a high-degree node, and the two nodes it brings together are drawn from its own neighborhood, which is better connected than the network at large. So closure tends to wire together nodes of *similar*, and similarly high, degree. The very mechanism that the disassortative models of the other chapters lack — linking within a neighborhood rather than reaching out to a random or peripheral node — is what tips the correlations positive. This is the deeper lesson of the chapter, and of the review that develops it (Vazquez, 2002): the *sign* of the degree correlations is not an arbitrary fitting knob but a direct readout of the microscopic rule. Search and duplication reach outward, to the periphery, and so make hubs that shun one another; triadic closure works inward, within neighborhoods, and so makes hubs that keep each other's company.

3.6 Summary

Modeling social growth by triadic closure — a newcomer attaches to a node and then, over time, to that node's friends — and formalizing "friend of a friend" as a *potential link*, we obtain a tractable theory in which potential links are created in proportion to degree. From that one ingredient follow an effective "rich get richer," a power-law degree distribution with a tunable exponent $\gamma = 1 + 1/\beta$, a clustering hierarchy in which cliquishness falls as one over the degree, and — distinctively — assortative degree correlations. Connecting friends-of-friends thus reproduces, from purely local moves, the characteristic shape of social networks, and explains why the sign of their correlations differs from that of the technological and biological networks treated elsewhere in this book.

Chapter 4

Gene duplication

4.1 Networks that copy themselves

The two rules of the preceding chapters each had a newcomer reach into the existing network and *choose* where to attach — by surfing, or by closing a triangle. Biology offers a third rule, and in some ways a more primitive one: the network grows by *copying* a node that is already there. The newcomer does not pick its neighbors at all. It inherits them, wholesale, from a parent.

The setting is the inner machinery of the living cell. Genomes evolve in good part through *duplication* — a gene, or sometimes an entire genome, is accidentally copied — followed by *divergence*, the slow accumulation of mutations that make the two copies drift apart. That duplication is a major engine of evolution was argued classically by Ohno (1970). To turn this into a network statement we use the *protein interaction network*: each node is a protein (the product of a gene), and a link joins two proteins that physically interact — that bind to each other to do their job. Now watch what a gene duplication does to this network. Right after the gene is copied, the cell makes two identical proteins, so the new protein interacts with *exactly the same partners* as the old one: the duplicate node arrives already wired to all the neighbors of its ancestor (Figure 4.1). Mutations then add a few interactions and, more often, remove some, letting the twins diverge (Wagner, 2001; Solé et al., 2002). Both



(a) duplication (b) divergence

Figure 4.1: **Duplication and divergence.** (a) A gene duplication adds a copy i' (gray) wired to every interaction partner of the ancestor i, plus a link between the twins when the protein is self-interacting. (b) Divergence then prunes the now-redundant interactions (dotted), each partner kept by only one twin, so the copies drift apart.

steps are thoroughly local — the copy needs to know only its parent and its parent's neighbors, never the rest of the genome.

The same picture reaches beyond biology. A new web page is often made by copying another page's hyperlinks and then editing the list — duplication as copying, divergence as editing. Following Vazquez (2003) and Vazquez et al. (2003), this chapter shows that copy-and-diverge, like the rules before it, produces an effective "rich get richer," a power-law degree distribution, the clustering hierarchy, and — as actually measured in protein networks (Maslov and Sneppen, 2002) — *disassortative* correlations. It also reveals something new: a *multifractal* degree distribution, which we will unpack when we reach it, a fingerprint left by the act of inheritance.

4.2 Duplication and divergence, in their simplest form

Start with the simplest version, treating duplication and divergence as independent random events. The single idea that drives everything is this: a node gains a new link whenever one of its *neighbors* is duplicated, because the neighbor's copy inherits a link back to it. A well-connected node has many neighbors, so it has many chances

to be on the receiving end of a duplication — and there, already, is the rich getting richer. The box makes this precise and reads off the degree distribution; the conclusion follows it.

The mathematics behind it: degree growth and the power law from duplication *(this box can be skipped)*

In a mean-field (average-quantities) description, with increasing the number of nodes n the degree of node i evolves as

$$\frac{\partial k_i}{\partial n} = \nu_D\, k_i + \nu_C\,(n - k_i) - \nu_L\, k_i, \tag{4.1}$$

with ν_D, ν_C, ν_L the per-node rates of duplication, random link creation, and link loss. The first term is the key one: it is proportional to k_i because each of i's k_i neighbors, if duplicated, hands i a new link. Each duplication adds one node, so $\nu_D = 1/n$; writing $\nu_C = \mu_0/n$ and $\nu_L = \mu_1/n$ reduces the equation to

$$n\,\frac{\partial k_i}{\partial n} = \mu_0 + (1 - \mu_1)\, k_i, \tag{4.2}$$

whose growth rate is linear in the degree. Integrating it gives the degree of node i as the network grows,

$$k_i(n) = \left(k_i(n_i) + \frac{\mu_0}{1-\mu_1}\right)\left(\frac{n}{n_i}\right)^{\beta} - \frac{\mu_0}{1-\mu_1}, \tag{4.3}$$

where n_i and $k_i(n_i)$ are the network size and the node's degree at the moment it was added, and the exponent is

$$\beta = 1 - \mu_1 \qquad (\mu_1 < 1), \tag{4.4}$$

the condition $\mu_1 < 1$ keeping the network from emptying out. To turn this growth law into a degree distribution, note that node i has degree above k exactly when its age and birth degree make Eq. (4.3) exceed k,

$$P(k_i > k) = \Theta\left[\left(k_i(n_i) + \frac{\mu_0}{1-\mu_1}\right)\left(\frac{n}{n_i}\right)^{\beta} - \frac{\mu_0}{1-\mu_1} > k\right], \tag{4.5}$$

with $\Theta[\cdot]$ the step function, 1 when its bracket holds and 0 otherwise. Both its birth size n_i and its birth degree $k_i(n_i)$ are random: for nodes added at a constant rate n_i is uniform, $P(n_i{=}n) = 1/n$, while

a node inherits its ancestor's degree, so $P[k_i(n_i){=}k'] = p_{k'}$ in the stationary state. Averaging over both,

$$P(k_i > k) = \sum_{k'} p_{k'} \int_1^n \frac{dn_i}{n}\, \Theta\left[\left(k' + \frac{\mu_0}{1-\mu_1}\right)\left(\frac{n}{n_i}\right)^{\beta} - \frac{\mu_0}{1-\mu_1} > k\right] \tag{4.6}$$

Differentiating this cumulative distribution and taking $n \gg 1$ leaves, for nodes added at a constant rate, a power-law degree distribution

$$p_k = \frac{\partial P(k_i > k)}{\partial k} \sim \left(\frac{\mu_0}{1-\mu_1} + k\right)^{-\gamma}, \qquad \gamma = 1 + \frac{1}{1-\mu_1}. \tag{4.7}$$

The result is once again a power law, Eq. (4.7), and its exponent is controlled entirely by μ_1, the rate at which divergence removes links: the more aggressively the duplicates shed interactions, the steeper the tail and the fewer the giant hubs. As in the previous two chapters, the power law can be traced to a single term that grows in proportion to degree — here the duplication term of Eq. (4.2) — and without it the distribution would decay as a featureless exponential. Copying, all by itself, is a rich-get-richer engine.

4.3 Cliquishness from self-interaction

If divergence added and removed links purely at random, this model would make no triangles, and its cliquishness would fade to nothing in a large network. But biology supplies one specific local source of triangles. Some proteins are *self-interacting* — they bind to copies of themselves. When the gene for such a protein is duplicated, the ancestor and its new twin can bind to *each other*, because each still carries the self-binding site (Wagner, 2001). That single interaction, laid down at the moment of duplication, is the seed of a triangle. The short box turns this into the clustering law.

The mathematics behind it: the clustering hierarchy from self-interaction *(this box can be skipped)*

Let a duplicated self-interacting protein bond to its ancestor with probability q_v. Then whenever a neighbor of node i is duplicated, i gains not only the copy as a new neighbor but, with probability q_v, a link between two of its neighbors (the ancestor and the copy). Hence the number of links e_i between the neighbors of node i change as

$$\frac{\partial e_i}{\partial n} \approx q_v \frac{\partial k_i}{\partial n}. \tag{4.8}$$

Integrating, assuming $e_i = 0$ when $k_i = 1$, we get $e_i = q_v(k_i - 1)$. Inserting this result into the clustering definition,

$$\langle c \rangle_k = \frac{2e(k)}{k(k-1)} \approx \frac{2q_v}{k}. \tag{4.9}$$

So cliquishness again falls as one over the degree, Eq. (4.9) — the clustering hierarchy of the introduction, reached now by a third independent route. There is one telling refinement: because removing an interaction also destroys triangles, and the biggest hubs have the most triangles to lose, divergence makes the clustering of high-degree nodes decay a little *faster* than one over the degree. That slightly-too-steep falloff is a small signature that distinguishes duplication from the search and closure mechanisms.

4.4 When duplication and divergence are coupled

In a real protein network the two processes cannot truly be independent. The network does a job — it keeps the cell alive — so losing the wrong interaction can be lethal, and which links a duplicate keeps is under natural selection. The classical view of Ohno (1970) lets one redundant copy decay freely; the more modern *subfunctionalization* picture of Force et al. (1999) has the two copies divide the ancestor's duties between them, each keeping part, together keeping the whole. Either way the fates of the two copies' links are *coupled*: a shared interaction is far more likely to be retained by one copy

than dropped by both. To capture this, Vazquez et al. (2003) use the following rule. Starting from two connected nodes, repeat:

- *Duplication.* Pick a node i at random and add a copy i' joined to every neighbor of i; with probability q_v also add the link i–i' (the self-interaction case).
- *Divergence.* For each node j now joined to both i and i', delete one of the two redundant links (i, j) or (i', j), at random, with probability $1 - q_e$.

The divergence step touches only the *redundant* pairs just created, and that is what couples it to duplication. Genomic data showing that duplicates diverge soon after they are born (Wagner, 2001) justify treating divergence as finishing before the next duplication, so the node count n doubles as a clock. A related model with lopsided divergence is studied in Vazquez (2003).

Coupling makes the large-scale behavior considerably richer, in two ways worth previewing before the box. First, a single number — the retention parameter q_e — decides whether the network stays sparse or grows ever denser, with a sharp switch between the two regimes. Second, the degree distribution is no longer a plain power law but a *multifractal* one. A quick word on that term: a plain power law is described by one exponent, the same at every scale; a multifractal is described by a whole continuum of exponents, because different "moments" of the distribution (the average degree, the average of the squared degree, and so on) scale with different powers of the network size. The box records the transition and the moment exponents; the interpretation follows.

The mathematics behind it: the density transition and the multifractal spectrum *(this box can be skipped)*

Tracking the mean degree through one duplication event gives

$$\langle k\rangle(n+1) = \frac{n\langle k\rangle(n) + 2q_v + (2q_e - 1)\langle k\rangle(n)}{n+1} \tag{4.10}$$

which in the large-network limit switches behavior at $q_e = 1/2$: for $q_e < 1/2$ the mean degree saturates at the finite value $\langle k\rangle = 2q_v/(1 - 2q_e)$, while for $q_e > 1/2$ it grows without bound as $N^{2q_e - 1}$. A rate-equation treatment in the manner of Kim et al. (2002) gives

an attachment rate $A(k) = (q_v + q_e k)/n$ — linear in degree again — and predicts that the l-th moment of the degree distribution scales as $M_l \sim n^{\sigma_l(q_e)}$ with

$$\sigma_l(q_e) = l\, q_e + 2\left[\left(\frac{1+q_e}{2}\right)^l - 1\right]. \tag{4.11}$$

For each l there is a value q_l at which σ_l changes sign, with $q_1 = 1/2$ and $q_2 = 2\sqrt{3} - 3 \approx 0.46$.

The two messages of the box are these. First, Eq. (4.10) shows a genuine switch at $q_e = 1/2$: keep too few links and the network settles into a sparse steady state; keep enough and it densifies forever. Second, and more subtle, the moment exponents σ_l in Eq. (4.11) depend on l in a curved, *nonlinear* way — the hallmark of a multifractal. Where the earlier chapters gave clean single power laws, duplication gives a spectrum of exponents, and the reason is exactly the inheritance at its core: each newcomer copies the degree environment of its parent, injecting a kind of heritable heterogeneity that a single exponent cannot capture. The deletion of links from the copies is what converts that inheritance into multifractality.

4.5 Disassortative correlations

Simulations spanning network sizes from a thousand to a million nodes confirm the analysis (Vazquez et al., 2003; Vazquez, 2003): the moment exponents match the multifractal prediction; the clustering falls with degree as a power law with exponent at or above one (steepening as link loss grows, just as Eq. (4.9) and its caveat foretold); and the average neighbor degree *decreases* with degree, $\langle k_{nn}\rangle(k) \sim k^{-0.1}$. The duplication network is therefore *disassortative* — its hubs are surrounded by poorly-connected partners rather than by each other.

This is the correct sign. Disassortative mixing is precisely what Maslov and Sneppen (2002) measured in the protein interaction network of yeast, and it files duplication alongside the technological networks of Chapter 2 and against the social networks of Chapter 3. With this third mechanism in hand, the three rules of the book so

far bracket the empirical landscape, as the table summarizes (here β is the rate at which cliquishness falls with degree, and α the slope of the neighbor-degree curve — positive for assortative, negative for disassortative):

Mechanism	$\langle c \rangle_k \sim k^{-\beta}$	$\langle k_{nn} \rangle_k \sim k^{\alpha}$
Connecting neighbors	$0 < \beta < 1$	$\alpha > 0$
Random walk	$\beta = 1$	$\alpha \leq 0$
Duplication–divergence	$\beta \geq 1$	$\alpha < 0$

Every local rule produces a clustering hierarchy; what differs is the *sign* of the correlations, which sorts the rules into the assortative (social, closure-driven) and disassortative (technological and biological, reach-driven) classes seen in real data.

4.6 The unifying principle

Three different stories — surfing a network, closing a triangle, copying a node — have each delivered the same effective "rich get richer." That is no accident, and the duplication picture makes the reason simple enough to state in a single sentence.

Picking a node *at random* plays no favorites: every node is equally likely, hub or not. But none of the local rules in this book act on a randomly chosen node. They act on a *neighbor* of one — the page a surfer steps to, the friend a friend introduces, the partner a duplicated gene inherits. And the chance that a particular node i is the neighbor of a randomly chosen node is

$$\frac{k_i}{\sum_j k_j}, \tag{4.12}$$

which is exactly the linear preferential-attachment probability of the Barabási–Albert model (Barabási and Albert, 1999). High-degree nodes are over-represented among the neighbors of a random node for the most elementary of reasons: they belong to more neighborhoods, so more random nodes have them as a neighbor. Preferential attachment never has to be imposed. It is induced automatically the instant a growth rule takes one step out from a random node to one of its neighbors. The clustering hierarchy has

the same local root — a rule that makes one link to a node tends to make another among that node's neighbors — and the sign of the degree correlations depends on *which* neighbor gets chosen. Locality is the common cause, and the three mechanisms are three faces of it.

4.7 Summary

Biological networks grow by duplication: a node is copied along with its connections, and the copies then diverge by gaining and losing links. Because a node gains a link whenever one of its neighbors is duplicated, the rule builds in an effective "rich get richer" and hence a power-law degree distribution, while duplications of self-interacting proteins generate a clustering hierarchy that link loss steepens. Coupling divergence to duplication adds a density transition at $q_e = 1/2$ and a multifractal degree distribution born of inheritance, and the model reproduces the disassortative correlations measured in real protein networks. Together with the two previous chapters, duplication completes a simple and general principle: a growth rule that attaches to a *neighbor* of a randomly chosen node attaches preferentially by degree — and so preferential attachment, the clustering hierarchy, and degree correlations are all the generic harvest of locality.

Chapter 5

Activity specialization

5.1 Projects as growing networks

The duplication rule of the previous chapter was born in biology, but copying is not a biological privilege. This chapter applies the same idea to a very different system — the schedule of a large engineering project — and gets, in return, a surprising practical payoff: an explanation of why big projects so reliably run late.

A project schedule can be drawn as a network, called an *activity network*. Its nodes are the individual activities, or tasks, that make up the project, and its links record *precedence*: a link from activity a to activity b means b cannot begin until a is finished — you cannot paint a wall before it is built. Because work only flows forward in time, the network has no loops; in the jargon it is a *directed acyclic graph*, with starting activities (no predecessors) and finishing activities (no successors). For such a network there is one object that has ruled project management since the 1950s: the *critical path*. This is the longest chain of dependent activities running from start to finish — the route through the project with no slack anywhere along it, so that delaying *any* activity on it pushes back the whole project's end date by the same amount (Kelley and Walker, 1959). The manager's time-honored prescription is therefore simple: watch the critical path, and guard it at all costs.

And yet projects keep finishing late. A survey of more than ten

thousand of them found only a few percent delivered on time and on budget, and fifteen years of improvements in management practice barely moved that figure (Flyvbjerg et al., 2003). This stubborn record hints that the critical path — a single one-dimensional thread — may be the wrong thing to watch in schedules that have grown into genuinely complex networks. Following Vazquez et al. (2023), this chapter grows such schedules from a local rule, shows that the critical path *shrinks toward irrelevance* as projects get bigger, and relocates the real source of delay risk in the network as a whole.

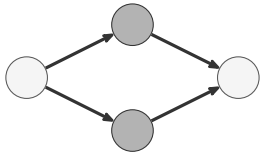
5.2 The duplication–split rule

Where does a schedule come from? Not in one piece. A plan is *refined*: a coarse activity is broken into finer ones, again and again, until the plan is detailed enough to execute. There are two natural ways to break an activity in two, and they correspond to two kinds of work. This is our local rule, with a single dial, the duplication probability q. Beginning from two activities in series, repeat: pick an existing activity at random and, with probability q, *duplicate* it, otherwise *split* it.

- *Duplication* (probability q). A *generic* activity is split into two smaller activities that run *in parallel* — think of doubling a crew so two teams do the same kind of work at once. Both copies inherit *all* the predecessors and *all* the successors of the parent. Duplication *widens* the schedule, adding parallel strands of work.

- *Split* (probability $1 - q$). A *specialized* activity is cut into two activities run *in sequence* — one specialist does the first part and hands off to another who does the second. The parent keeps its predecessors, the new activity takes over its successors, and a link joins the two. Splitting *lengthens* the schedule, extending a chain.

So the dial q measures how a project is organized (Figure 5.1). Small q means almost every refinement is a split: work is broken into ever-longer chains of specialists, and the schedule is nearly a single long line. Large q means almost every refinement is a duplication:



(a) duplication: in parallel



(b) split: in sequence

Figure 5.1: **Refining an activity.** A generic activity is replaced either by (a) two copies that run *in parallel*, both inheriting the predecessor (left, gray) and successor (right, gray) of the parent — duplication, which widens the schedule — or by (b) two specialized activities that run *in sequence* — a split, which lengthens it.

generic work is parallelized into many simultaneous strands, and the schedule fans out into a broad web. The same copying that duplicated a gene in the previous chapter here duplicates an activity; the new ingredient is the *split* — the act of *specialization* that gives the chapter its name — and the forward-flowing, start-to-finish structure of a schedule.

5.3 Predecessors and successors: identical power laws

Because the rule copies an activity together with its predecessors and successors, it is a duplication rule of the kind that produced power-law degrees in the previous chapter — only now there are two degrees to track, the number of predecessors (the in-degree) and the number of successors (the out-degree). The box works out the distribution of predecessors by following how many activities have each number of them; the same argument applies verbatim to successors. The result is in the paragraph after.

The mathematics behind it: the distribution of predecessors *(this box can be skipped)*

Let $n_k(n)$ be the number of activities with k predecessors when the schedule has n activities in all. Adding one activity changes this count by

$$n_k(n+1) = n_k(n) + q\left[\frac{k-1}{n}\,n_{k-1}(n) - \frac{k}{n}\,n_k(n) + \frac{1}{n}\,n_k(n)\right] + (1-q)\,\delta_{k,1} \tag{5.1}$$

the bracket collecting the duplication events (an activity climbs from $k-1$ to k predecessors when one of its predecessors is duplicated, and so on) and the last term adding a fresh single-predecessor activity at each split. Looking for a stable shape $n_k(n) = n\,p_k$ turns this into a recurrence,

$$p_k = q\big[(k-1)\,p_{k-1} - k\,p_k + p_k\big] + (1-q)\,\delta_{k,1}, \tag{5.2}$$

which solves in closed form (with Γ the gamma function) to

$$p_k = \prod_{s=1}^{k-1} \frac{s}{\frac{1}{q}+s}\;p_1 = \Gamma\!\left(\tfrac{1}{q}+1\right)\frac{\Gamma(k)}{\Gamma\!\left(\tfrac{1}{q}+k\right)}\,p_1, \tag{5.3}$$

with the power-law tail

$$p_k \sim k^{-1/q}. \tag{5.4}$$

The result, Eq. (5.4), is a power law with exponent $\gamma = 1/q$. A wide, duplication-heavy schedule (large q) has the heaviest tail of highly-connected activities; at $q = 1/2$ we recover the familiar value $\gamma = 2$. The very same reasoning, run on successors instead of predecessors, gives the *identical* distribution for the out-degree. That coincidence is the model's signature prediction — the spread of predecessor counts and the spread of successor counts should look the same — and it follows directly from the even-handed way duplication treats the two ends of an activity.

5.4 The vanishing critical path

The degrees are a local property; the critical path is a global one. Remarkably, the same rule fixes how the critical path grows with the

size of the project, and the answer overturns the manager's intuition. The box does the short calculation; the consequences follow.

The mathematics behind it: how the critical path scales *(this box can be skipped)*

Let c be the number of activities on the critical path of an n-activity schedule. The path can only lengthen when a *split* lands on it (a split inserts an activity in series), and the chance that the activity picked for refinement lies on the critical path is c/n. Hence

$$\frac{dc}{dn} = (1-q)\,\frac{c}{n}, \tag{5.5}$$

which integrates to $c \sim n^{1-q}$, so the *fraction* of activities on the critical path shrinks as

$$\frac{c}{n} \sim n^{-\alpha(q)}, \qquad \alpha(q) \le q. \tag{5.6}$$

Because distances here grow as a power of size, $c \sim n^{1/D}$, the network is a *fractal* with dimension

$$D = \frac{1}{1-\alpha(q)}. \tag{5.7}$$

The headline is Eq. (5.6): for any amount of duplication ($q > 0$), the critical path is a *vanishing fraction* of the project as the project grows. The bigger and more parallel the schedule, the smaller the slice of the work that the manager's cherished path actually covers — so watching the critical path means watching an ever-shrinking sliver of the whole. (Simulations confirm the power-law shrinkage, with the measured exponent α a little below the simple bound q; for instance $\alpha \approx 0.11$, 0.28, 0.43 at $q = 0.20$, 0.50, 0.70.)

The scaling also tells us what *kind* of network a schedule is. In a small-world network, distances grow only logarithmically with size — doubling the network barely lengthens the longest path. Here distances grow as a *power* of size, the mark of a *fractal* network: one whose "size" and "length" are tied together by a fixed dimension D, much as a physical fractal looks the same at every magnification. Duplication–split schedules are thus a class apart — power-law degrees like a scale-free network, but power-law (not logarithmic)

distances, unlike a small world. Specialization builds the long fractal chains; duplication stacks them up in parallel.

5.5 Real project schedules

This is not just a story about a model. Vazquez et al. (2023) examined 77 real construction projects, together worth more than ten billion dollars, which yielded 323 activity networks — snapshots of projects at different stages — ranging from about a hundred to sixteen thousand activities. Three predictions survive the encounter with data. First, the predecessor and successor counts are each power-law distributed, and fitting them *separately* yields duplication indices q that agree, just as the model demands they should. Second, those indices fall between roughly 0.1 and 0.5, most above 0.2: real schedules sit on the split-dominated, nearly-linear side — but with enough duplication to matter. Third, the share of activities on the critical path falls with project size right across the collection, confirming that the critical path really does dwindle as projects grow.

5.6 Delay risk as percolation

If the critical path fades to insignificance and yet projects are still late, where does the delay risk live? The answer is that it is spread across the whole network, and it is governed by *percolation* — the same mathematics that describes how water seeps through porous rock or how a forest fire spreads from tree to tree.

Picture an outside shock — bad weather, a strike, a pandemic — that pushes some activities past their planned finish. A delay only passes to a downstream activity if it is large enough to use up the spare time, the *free float*, built into that dependency; otherwise the buffer absorbs it and the delay stops there. Now treat each dependency as a channel that passes a delay along with some probability p, and ask what fraction of activities end up delayed beyond their own slack — the *delay incidence*. This is exactly a percolation problem: delays spread through the network of dependencies like fire through a forest, and percolation theory says such spreading has a sharp

threshold (Albert et al., 2000; Ellinas, 2019; Schwartz et al., 2002),

$$p_c = \frac{1}{\langle k \rangle}, \tag{5.8}$$

where $\langle k \rangle$ is the average number of downstream dependencies per activity. The meaning of a threshold is stark: below p_c delays stay local and project-wide overruns essentially never happen; above p_c a finite fraction of projects suffer a system-wide cascade — the fire jumps the firebreaks.

The data choose between two stories cleanly. If the critical path governed delay, the incidence ought to track $p\,c$ (transmission times critical-path size); in the real projects that correlation is negligible. If percolation governs delay, the incidence ought to switch on exactly at $p - p_c = 0$; and that is what the construction projects show — below threshold, delay incidence stays under one percent; above it, the incidence climbs, reaching fifteen percent of all activities in the worst cases. The true control knob of project delay is $p - p_c$, a property of the *whole* network, not $p\,c$, a property of one path through it.

Here is the chapter's practical lesson, and its link to the rest of the book. The very same local rule that sets the degree distributions also sets the average degree $\langle k \rangle$, and through Eq. (5.8) the threshold for catastrophic delay. A project's tendency to run late is written into the network its activities have grown into — not into any single path through it.

5.7 Summary

Project schedules are forward-flowing networks of activities, grown by a local refinement rule: pick an activity and either duplicate it into parallel copies (probability q) or split it into specialized sequential parts. This duplication–split rule predicts identical power-law distributions for the numbers of predecessors and successors, $p_k \sim k^{-1/q}$; a critical path that grows only as $c \sim n^{1-q}$ and so vanishes as a fraction of large projects; and a fractal, rather than small-world, geometry. All three hold in 77 real construction projects. And because the critical path becomes irrelevant at scale, delay risk is

better read through percolation, with a threshold $p_c = 1/\langle k \rangle$ fixed by the grown network's average degree — so that project performance, in the end, is a property of the activity network as a whole.

Chapter 6

Emergence of communities

6.1 Communities nobody designed

So far the local rules of this book have explained the *degree* structure of networks — their power laws, their clustering hierarchy, their correlations. This chapter turns to a feature of an altogether different kind: *community structure*. A community is a group of nodes that are densely connected among themselves but only sparsely connected to the rest of the network — a clump that hangs together. They are everywhere in real data: people fall into social circles, web pages into topics, proteins into functional modules, each group talking mostly to itself. Splitting a network into such groups is what researchers mean by *community detection*, and a network that splits cleanly is said to have strong community structure.

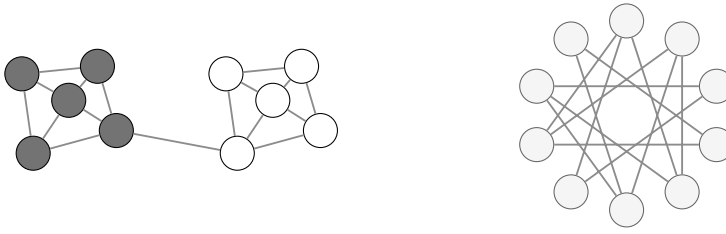
Where do these communities come from? The usual answer points to differences between the nodes themselves — *node heterogeneity*, in the jargon. People of the same profession or politics, web pages on the same subject, proteins with the same biological function: nodes of a kind, the story goes, are simply more likely to link to one another, and that built-in preference is what carves the network into groups. On this view, communities are stamped onto the network by properties the nodes carried in from the start.

The thesis of this chapter, following Vazquez (2025), is that no such stamp is needed. *Communities emerge from local growth rules even when every node is identical.* The first hint is decades old: Jin et al. (2001) showed that a social network grown purely by triadic closure — the friends-of-friends rule of Chapter 3 — develops communities although its nodes are all of one type, and the same was later seen in the gene-duplication models of Chapter 4 (Solé et al., 2002). If a featureless local rule segregates a network into groups all by itself, then community structure joins preferential attachment and the clustering hierarchy as one more thing that *emerges* from locality rather than being designed in.

To make that claim sharp we should ask it as a quantitative question, and the right way to frame it comes from a corner of mathematics called *Ramsey theory.* Ramsey theory studies how large a random structure must become before some orderly pattern is forced to appear. Its best-known nugget is a party fact: in any gathering of six people, there are always either three who all know one another or three who are all mutual strangers — order of that kind is unavoidable once the group is big enough. We will ask the analogous question of growing networks: given a local growth rule, how large must the network become before it is virtually *certain* to contain communities? This chapter is, in effect, a Ramsey theory of community structure in growing networks.

6.2 The Ramsey community number

Before we can measure when communities appear, we need a trustworthy way to *detect* them, and here some care is required, because the most popular method can be fooled. That method, *modularity optimization*, looks for the division of a network that packs the most links inside groups; the trouble is that it will happily report groups even in a completely random network that has no real structure at all (Newman, 2006; Fortunato, 2010). To avoid being deceived, the work relies instead on two more discerning tools: the *stochastic block model*, which fits the network to a statistical model of blocks and keeps only the structure that genuinely earns its keep (Karrer and Newman, 2011), cross-checked against *Infomap*, which finds groups by watching where a random walker gets trapped (Rosvall



(a) grown by a local rule (b) after random rewiring

Figure 6.1: **Communities emerge from local rules.** (a) A network grown by a local rule settles into dense communities (two here, one dark gray, one white) joined by only a few links. (b) Shuffling the links at random while keeping every node's number of connections destroys the communities — so the structure was the work of the rule, not of the degrees alone.

and Bergstrom, 2008). Both, crucially, report *no* communities in a truly random network.

That last point also supplies the essential control. To be sure a community is the work of the growth rule and not merely of the degrees, one compares the grown network against a *rewired* version of itself — the links shuffled at random while every node keeps its number of connections (this shuffling is called the configuration model). If the communities survive the shuffle, they were never more than an accident of the degrees; if they vanish, they were the doing of the local rule. For every local model in this chapter, the shuffled version has no communities at all — so whatever structure we find is genuinely built by the dynamics.

With detection settled, we can pose the question precisely (Figure 6.1). The box below states the two definitions that do the work; both are short, and their meaning is given in words right after, so the box can be skipped.

The mathematics behind it: the communities likelihood and the Ramsey community number *(this box can be skipped)*

For a growth rule f_G that produces a network of n nodes, let a detection method f_κ return the number κ of communities it finds. Define the *communities likelihood* as the probability — estimated by generating many networks — that an n-node instance has more than one community,

$$P_\kappa(n) = \text{Prob}\{\, \kappa \geq 2 \,\}. \tag{6.1}$$

The *Ramsey community number* is then the smallest size at which communities are all but guaranteed, at a confidence $1 - \epsilon$ (we use $\epsilon = 0.05$, i.e. 95%):

$$r_\kappa = \min\{\, n : P_\kappa(n) \geq 1 - \epsilon \,\}. \tag{6.2}$$

A rule is said to have the *emergent communities property* when such an r_κ exists.

In plain terms: $P_\kappa(n)$ is the chance that a network grown to size n has communities, and the Ramsey community number r_κ is the size beyond which that chance is essentially $(1 - \epsilon)100\%$. A *finite* r_κ is the fingerprint of emergence — grow the network past it and communities are a near-certainty. A rule for which no such size exists, no matter how large the network, simply does not make communities on its own. The whole question "do communities emerge from this rule?" thus reduces to "is r_κ finite?"

One caution should be lodged now and collected later. The definition above is stated *relative to a detection method* f_κ: r_κ is a property of the network *and* of the rule used to interrogate it, and the two cannot be separated. For most of this chapter that dependence is a technicality, the choice of method shifting the numbers without touching the conclusions. In the last part of the chapter, where we can compute r_κ exactly, it will turn out to be rather more than a technicality.

6.3 Local rules grow communities

We now line up two local rules against two controls. The local rules — local search and duplication–split, both met earlier in this book —

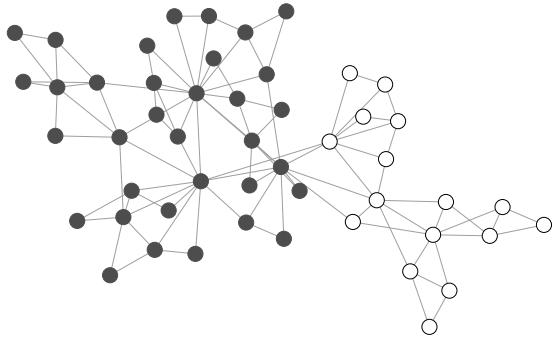


Figure 6.2: **Communities in a local-search network.** An instance of the local-search model LS(n=50, d=1). The stochastic block model resolves two communities, shown dark gray and white; they emerge even though the rule treats every node alike and never mentions groups.

have the emergent communities property; the controls — reshuffling a network's links, and growing one with no locality at all — do not. Holding the four side by side is the whole argument of the chapter in miniature.

Local search. The model LS(n, d) is the search rule of Chapter 2 in undirected form (Vazquez, 2001): each new node takes a short walk of d steps from a random starting node and links to every node it visits. It carries preferential attachment and high clustering, and — though nothing in the rule ever mentions groups — the block model resolves clear communities, with P_κ climbing from near zero around $n = 10$ to a near-certain one beyond about $n = 100$, and staying there out to a hundred thousand nodes (Vazquez, 2025). Figure 6.2 shows one such network.

Duplication–split. The model DS(n, q) is the undirected cousin of the project-schedule rule of Chapter 5: a new node either copies all of a random node's links (probability q) or splits one of its links in two. This model is the decisive case, because it builds *no triangles whatsoever* — duplication makes none, and splitting destroys any

that might form — and yet communities emerge just as reliably. The lesson is sharp: a network does not need to be cliquish to have communities. What it needs is locality, not triangles. Figure 6.3 shows an instance: two communities, and not a triangle in sight.

Degree-preserving randomization. To be sure the communities are the work of the local rule and not of the degrees alone, we need a way to wipe out the local structure while leaving every degree untouched. The *configuration model*, met when we defined the Ramsey number, does exactly that: it takes a finished network and reshuffles its links at random, keeping each node's number of connections fixed but scrambling *which* nodes are joined. Applied to a local-search or a duplication–split network, this reshuffling destroys the communities outright — the randomized versions have no finite Ramsey number at all (the two "randomized" rows in the table below). Whatever the local rules built, the bare list of degrees cannot sustain.

Barabási–Albert. Finally, BA(n, m) — the canonical preferential-attachment model — is our *non-local* control: a growing, rich-get-richer network in which each newcomer attaches its m links to nodes drawn from the *whole* graph, never once consulting a neighborhood. Despite its growth and its rich-get-richer rule it has no emergent communities (the last row of the table), and the next section is devoted to teasing apart exactly why.

A few representative Ramsey numbers, based on numerical estimates for $\epsilon = 0.05$ (Vazquez, 2025), make the pattern concrete:

Model	r_κ (block model)	r_κ (Infomap)
LS$(d = 1)$	81	222
DS$(q = 0.5)$	98	622
LS$(d = 1)$, randomized	—	—
DS$(q = 0.5)$, randomized	—	—
BA$(m = 2)$	—	—

Every genuinely local rule has a finite Ramsey number — grow it past a few hundred nodes and communities are certain. The rules without local structure — the bottom three rows, the two reshuffled

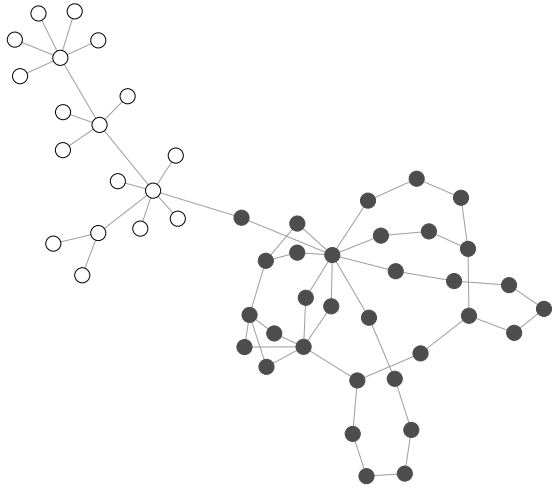


Figure 6.3: **Communities in a duplication–split network.** An instance of the duplication–split model DS(n=50, q=0.3). Two communities (dark gray and white) again emerge — even though this rule builds no triangles at all, so the network is far less cliquish than the local-search one of Figure 6.2.

networks and the Barabási–Albert model — have none: the dashes mean communities never become certain, however large the network. The exact figures differ between the two detection methods — there is no single "true" definition of a community, and where the line falls depends a little on how one draws it — but the *split* between local rules (finite) and non-local ones (infinite) is the same either way. Local structure is a necessary condition for communities to emerge.

6.4 When globality fails: the Barabási–Albert model

The natural control is the Barabási–Albert model (Barabási and Albert, 1999), the textbook growing network with *no* locality at

all: each newcomer attaches its m links to nodes chosen from the *whole* network in proportion to their degree. It has growth and preferential attachment, but no step that ever looks at a node's neighborhood — and it does *not* have the emergent communities property. Its community likelihood rises only to about one-half near a thousand nodes and then *falls* as the network grows further, never reaching certainty. The little structure it has comes from the mere age-ordering of its nodes, not from any local rule, and it washes out at scale.

The boundary case is illuminating. Give the model a baseline "attractiveness" a, so that a node is chosen with a probability proportional to $a + k_i / \sum_j k_j$ — a fixed head-start a on top of the usual degree preference. This tunes the degree exponent to $\gamma = 2 + a/m$. Tracking the Ramsey number across these variants reveals a clean rule of thumb: communities emerge exactly when the network is *clustered* enough. The Ramsey number falls as the average clustering coefficient rises, and *diverges* — communities never become certain — as the clustering goes to zero (Vazquez, 2025). The standard Barabási–Albert model sits right at that vanishing point, because its clustering decays toward zero as it grows (like $(\ln n)^2/n$): the bigger it gets, the fewer short cycles it has, and short cycles are what local structure is made of. So even here the principle holds — where the dynamics happen to generate appreciable clustering (an *implicit* bit of local structure), communities appear; where clustering fades, they do not.

6.5 How local must "local" be?

If locality is the cause, exactly how local must a connection be to count? Two dials settle it: the length of the cycles a rule creates, and the distance it reaches across. Triadic closure closes triangles (cycles of length three); the duplication–split rule works with squares (length four). To sweep the cycle length freely, use the *bubble model*: at each step a chain of L new nodes is added and its two ends are attached to the two ends of a randomly chosen existing link, creating a fresh cycle of length $L + 2$. As L grows — as the rule weaves ever longer cycles — the Ramsey number climbs with it, and steeply: from $r_\kappa = 108$ for the shortest cycles to more than fifty thousand

by $L = 9$ (Vazquez, 2025), as the table below records. The climb is not quite smooth — riding on top of it is a regular even–odd alternation, the odd-length cycles bringing communities on sooner than the even cycles around them, an effect plainest for the shorter cycles — but the overall message is unmistakable: short cycles, tight local feedback, bring communities on soonest.

L	cycle length $L+2$	r_κ
1	3	108
2	4	326
3	5	246
4	6	2201
5	7	752
6	8	6595
7	9	2740
8	10	15863
9	11	56160

Distance matters just as much. In the bubble model the chain's two ends attach to nodes that are a single step apart (the ends of one link). Loosen this so the ends may attach to nodes up to W steps apart, with the chain length fixed at $L = 1$. For $W = 1$ communities still emerge with certainty, at $r_\kappa = 108$; stretching to $W = 2$ pushes the threshold up roughly tenfold, to $r_\kappa = 1090$, but the communities still come. At $W = 3$ — and likewise if the ends are joined to nodes picked anywhere at random — they stop appearing altogether, the Ramsey number ceasing to exist (the dash in the table) (Vazquez, 2025).

W	r_κ
1	108
2	1090
3	—

For this model, "local" means roughly two steps: communities form when new links reinforce ties among nearby nodes, and dissolve once links are allowed to leap across the network. This is the whole book's premise made quantitative. It is not growth, nor preferential attachment, nor even clustering in the abstract that carves a network into communities — it is the *locality* of the rule by which it grew.

6.6 Networks we can solve exactly

Everything so far has been measured, not proved. The Ramsey numbers of the last three sections came from growing many networks and running a detection algorithm on each — honest work, but it leaves the central question in the dark. We know *that* communities appear beyond a certain size. We do not know *why*, and we cannot say what the size depends on.

To answer that we need cases simple enough to solve with pencil and paper. Three have recently been found, and the rest of this chapter is their story. Each strips the problem down to a different bone. The *ring* (Vazquez, 2026d) removes growth, randomness, and heterogeneity all at once, leaving nothing but local wiring. The *diamond lattice* (Vazquez, 2026c) — the same lattice whose shortest paths we shall count in Chapter 7 — lets us watch, exactly, why the evidence for communities accumulates at all. The *pseudofractal web* (Vazquez, 2026b) asks whether the communities we find are real or an illusion cast by the hubs.

A warning before we start: these three networks are not *grown* at all. They are fixed, deterministic constructions, and no detection algorithm is ever run on them. What they isolate is locality itself, stripped of every other ingredient, and what they answer is not "can an algorithm find the communities?" but "at what size does the split become the better description?" That is the question the Ramsey number was always asking, and here we can answer it exactly.

The trick that makes all three solvable is to use the block model not as an algorithm but as a *formula*. Given a network and a candidate split, the probability of the data under each hypothesis can be integrated in closed form, and the two compared.

The mathematics behind it: the block model as a formula
(this box can be skipped)

Suppose the nodes are divided into blocks, and that the chance of a link between two nodes depends only on which blocks they are in: θ_1 within block 1, θ_2 within block 2, θ_{12} between. We do not know these probabilities, so we average over them, giving each a Beta(α, α) prior, and likewise average over the label prior π that decides how

nodes are apportioned:

$$P = \frac{1}{Z} \int \prod_{i \in \{1,2\}} \theta_i^{E_i} (1-\theta_i)^{\binom{n_i}{2} - E_i} \times \theta_{12}^{E_{12}} (1-\theta_{12})^{n_1 n_2 - E_{12}} \times \pi^{n_1} (1-\pi)^{n-n_1} \, P(\theta) P(\pi) \, d\theta \, d\pi. \tag{6.3}$$

Here n_i is the size of block i, E_i the number of links inside it, and E_{12} the number crossing between. Every integral is of the same standard shape,

$$\int_0^1 x^{E+\alpha-1} (1-x)^{N-E+\alpha-1} \, dx = \mathrm{B}(E{+}\alpha,\, N{-}E{+}\alpha), \tag{6.4}$$

where B is the Beta function. So each hypothesis collapses to a product of Beta functions, and their ratio R — the *evidence ratio* — can be evaluated exactly. All we ever need is a handful of edge counts.

Writing R for the ratio of the evidence for the split to that for no split, the posterior weight of the partition is

$$P(\text{split}) = \frac{R}{1+R} = \frac{1}{1+e^{-\log R}}, \tag{6.5}$$

so $\log R > 0$ means the split is the better description, and the Ramsey number is the smallest size at which $P(\text{split})$ reaches the certainty q we demand.

6.7 The ring wants to be broken

Begin with the least promising network imaginable. Place n nodes on a circle and join each to its c nearest neighbors on either side — the *circulant ring* $C_n(1, \ldots, c)$, the unrewired lattice that Watts and Strogatz started from. Every node is identical. Every node has the same degree $2c$, and the ring has cn links. Nothing grows, nothing is random, and there is no group of nodes that belongs together more than any other. If any network has no communities, this one does not.

Ask the block model anyway. The candidate split is into two contiguous arcs, and the counts it needs are elementary.

The mathematics behind it: the ring's edge counts and its evidence ratio *(this box can be skipped)*

Take an arc of n_i consecutive nodes. A link of span k joins nodes k apart, and $n_i - k$ such links fit inside the arc, so summing over the spans $k = 1, \dots, c$,

$$E_i = \sum_{k=1}^{c} (n_i - k) = c\, n_i - \frac{c(c+1)}{2}, \tag{6.6}$$

and the pairs inside the arc that are *not* linked number $\binom{n_i}{2} - E_i = \binom{n_i - c}{2}$. Only links near the two cut points cross between the arcs, and counting them by span gives

$$E_{12} = 2 \sum_{k=1}^{c} k = c(c+1), \tag{6.7}$$

independent of n. As a check, $E_1 + E_2 + E_{12} = cn$. For the balanced split $n_1 = n_2 = m \equiv n/2$; the unpartitioned ring has all cn links among $\binom{n}{2}$ pairs. Feeding these into Eq. (6.3) and using Eq. (6.4) on every integral,

$$\begin{aligned} R = \Big[& \mathrm{B}\big(cm - \tfrac{c(c+1)}{2} + \alpha,\ \tbinom{m-c}{2} + \alpha\big)^2 \\ & \times \mathrm{B}\big(c(c+1) + \alpha,\ m^2 - c(c+1) + \alpha\big) \\ & \times \mathrm{B}(m + \alpha,\ m + \alpha) \Big] \\ \Big/ \Big[& \mathrm{B}(\alpha, \alpha)^2\, \mathrm{B}\big(cn + \alpha,\ \tbinom{n}{2} - cn + \alpha\big) \\ & \times \mathrm{B}(\alpha,\ n + \alpha) \Big]. \end{aligned} \tag{6.8}$$

No approximation has been made: this is exact for every n, c and α.

Evaluating Eq. (6.8) gives an answer that depends, with startling sharpness, on the range c (Figure 6.4a). The plain cycle, $c = 1$, is *never* split: its evidence ratio decays as a power of the size,

$$\log R \sim A - (2\alpha + 3) \log n \longrightarrow -\infty, \tag{6.9}$$

so that $P(\text{split}) \sim n^{-(2\alpha+3)}$ vanishes and $r_\kappa = \infty$. But the next-nearest-neighbor ring, $c = 2$, is split as soon as it exceeds about thirty-five nodes — $r_\kappa \simeq 34$ at 95% certainty — and beyond that the evidence piles up in proportion to the size, $\log R \sim (\ln 2)\, n$. A featureless ring of beads, given one extra reach, acquires communities.

Why should one extra neighbor make such a difference? The published answer is that a plain cycle offers no density contrast for the block model to exploit. That cannot be right: an arc of a cycle is twice as dense as the whole cycle, and so is an arc of any other ring — the contrast is a factor of two for *every* c. The real reason is a cancellation, and it is worth seeing because it generalizes.

The mathematics behind it: why the ring breaks, and when *(this box can be skipped)*

Two quantities compete. Cutting the ring into two arcs of $n/2$ makes each arc look twice as dense as the whole ring did: the arcs hold nearly all cn links but only about half the node pairs. Each link therefore gains $\ln 2$ of evidence, for a total *gain* of $cn \ln 2$. Against this stands the price of naming the split — of saying *which* nodes go in which block. That price is the label term of Eq. (6.8), and at $\alpha = 1$ it is not merely of order $n \ln 2$ but exactly a binomial coefficient. With $n = 2m$,

$$\frac{\mathrm{B}(m+1, m+1)}{\mathrm{B}(1, n+1)} = \frac{(m!)^2}{(2m+1)!}\,(n+1) = \frac{(m!)^2}{n!} = \binom{n}{n/2}^{-1}, \qquad (6.10)$$

so the label term contributes

$$-\log \binom{n}{n/2} \simeq -\,n \ln 2, \qquad (6.11)$$

the entropy of labelling n nodes with two labels. Subtracting the price from the gain,

$$\log R \;\simeq\; \big(cn - n\big) \ln 2 \;=\; (c-1)\, n \ln 2. \qquad (6.12)$$

Since cn is the number of links m_e, this reads $\log R \simeq (m_\mathrm{e} - n) \ln 2$: the ring is split exactly when its links outnumber its nodes.

Equation (6.12) goes beyond what the ring paper states — it

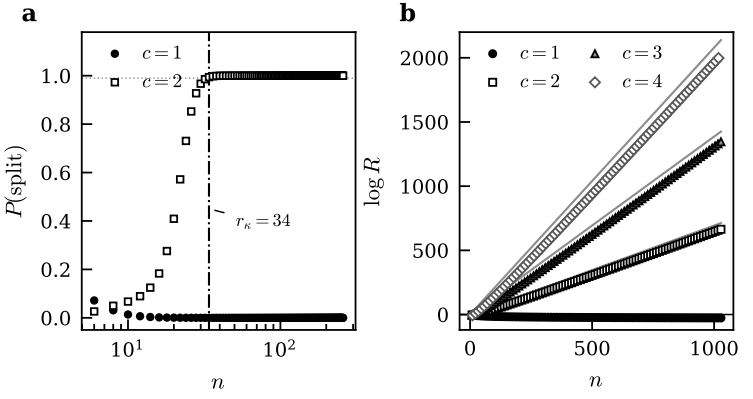


Figure 6.4: **The ring wants to be broken.** (a) The probability that a block model prefers two arcs to no split. The plain cycle ($c = 1$) is never split; the next-nearest-neighbor ring ($c = 2$) is split beyond $r_\kappa \simeq 34$ nodes (dash-dotted line; dotted line marks 99% certainty). (b) The evidence for the split against the prediction $\log R \simeq (c-1)\, n \ln 2$ (lines), Eq. (6.12), for $c = 1, \ldots, 4$. The cycle lies exactly on zero: it is balanced, not reluctant.

reports only the $c = 2$ case — but it is easily checked against the exact formula for $c = 1, \ldots, 4$, and it passes (Figure 6.4b). It also dissolves the mystery of the cycle. The cycle is not *reluctant* to break; it is exactly *balanced*. A cycle has precisely as many links as nodes, so gain and price cancel to the last term, and the fate of the split is left to the smaller corrections of Eq. (6.9), which happen to be negative. Move to $c = 2$ and the cancellation is broken by a whole factor of n. The celebrated contrast between the cycle and the ring is thus not a transition at all but a knife edge: $c = 1$ sits at the zero of a perfectly smooth family, and the evidence grows linearly in $c - 1$ thereafter.

And now the connection this book has been building toward. The quantity $m_\mathrm{e} - n$ is, up to one, the number of *independent cycles* a connected network carries. A tree has none, and is never split. A single ring has exactly one, and sits exactly on the fence. A network

with an extensive number of independent cycles is split with an evidence proportional to how many it has. The same short loops that Chapter 7 will show multiplying shortest paths are what carve a network into communities here — two faces, again, of the one local coin.

There remains something troubling about calling the two arcs communities, and it should be said plainly, because the paper does not. The ring looks the same from every node; rotate it by one bead and it is unchanged. So if cutting it at some point is favored, then cutting it at any *other* point is favored exactly as much: all n rotations of the balanced split have identical evidence. The block model is not telling us where the communities are. It is telling us that the ring *must be cut somewhere*, while remaining perfectly silent about where — and it is the ring's own symmetry that keeps it silent. Physicists know this situation well. A magnet below its critical temperature must point somewhere, though nothing in its physics prefers north to south; the symmetry of the laws survives in the multiplicity of the outcomes, not in any one of them. The ring's communities are of exactly this kind: real, in that the split genuinely describes the network better; degenerate, in that no particular split is preferred. The ring does not have communities so much as it wants to be broken.

6.8 Why communities emerge at all

The ring shows that locality suffices. It does not explain why size should be the deciding variable — why every one of these networks has a threshold at all. For that we turn to the diamond lattice, which we shall meet again in Chapter 7: begin with a single link between two poles A and B, and at each generation replace *every* link by b parallel paths of s links each. Take $b = s = 2$, the classic Migdal–Kadanoff cell (Migdal, 1975; Kadanoff, 1976). Generation t has

$$m = (bs)^t = 4^t \text{ links}, \qquad n_t = \tfrac{1}{3}\big(2 \cdot 4^t + 4\big) \text{ nodes}, \tag{6.13}$$

and a node born at generation k ends with degree $2 \cdot 2^{t-k}$, while the two poles are permanent hubs of degree $b^t = 2^t$. The lattice

is strongly degree-heterogeneous, which is what will make the next section's question sharp.

The b bundles hanging between the poles are the obvious candidate communities. Take block 1 to be one bundle together with the two poles, block 2 the other bundle. Because the bundles meet *only* at the poles, every crossing link joins a pole to a bundle-2 node, and the counts close in themselves.

The mathematics behind it: the counts, and the map that carries them *(this box can be skipped)*

For the bundle cut of the $b = s = 2$ diamond,

$$\begin{aligned} e_{11} &= \tfrac{4^t}{2}, & e_{22} &= \tfrac{4^t}{2} - 2^t, \\ e_{12} &= 2^t, & \kappa_1 &= 4^t + 2^t, \\ \kappa_2 &= 4^t - 2^t, \end{aligned} \tag{6.14}$$

where e_{rs} counts links between blocks r and s and κ_r is the sum of degrees in block r. The seam is *subextensive*: $e_{12} = 2^t = \sqrt{m}$, so the fraction of links crossing the cut decays geometrically, $e_{12}/m = (1/s)^t \to 0$.

The point of the lattice is that these tallies renormalize *exactly*. Collect them in $\mathbf{v} = (e_{11}, e_{22}, e_{12}, \kappa_1, \kappa_2)^\top$; then $\mathbf{v}(t{+}1) = M\mathbf{v}(t)$ with

$$M = \begin{pmatrix} 4 & 0 & 0 & 0 & 0 \\ 0 & 4 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & -2 & 4 & 0 \\ 0 & 0 & 2 & 0 & 4 \end{pmatrix}. \tag{6.15}$$

The entries read off the construction. A link inside either block is replaced by four links whose new nodes inherit its bundle label, so within-block links quadruple. A seam link joins a pole to a bundle-2 node; of its four children, the two at the pole stay seam links and the two at the far end are absorbed into block 2 — whence $e'_{12} = 2e_{12}$, with the displaced pair reappearing in e'_{22}. The spectrum is what matters:

$$\lambda_{\mathrm{vol}} = bs, \qquad \lambda_{\mathrm{seam}} = b. \tag{6.16}$$

Read backwards, M^{-1} is precisely the Migdal–Kadanoff decimation that strips the youngest generation; read forwards, M is the growth generator of the detection problem.

Two eigenvalues, and their ratio is the whole story:

$$\frac{\lambda_{\text{seam}}}{\lambda_{\text{vol}}} = \frac{1}{s}. \tag{6.17}$$

For $s \geq 2$ the seam is left behind by the bulk, so the boundary between the communities grows ever fainter relative to their interiors, and the split gets better and better with size. For $s = 1$ the two rates are equal, the seam keeps pace, and the lattice never separates — the diamond's exact counterpart of the plain cycle, and a pleasing echo of it. In a sentence: b sets how many communities there are, and s sets how sharply they are separated.

To weigh the evidence we need a null model that already knows the degrees — otherwise, as the next section will show, we would only be measuring the hubs. The degree-corrected block model (Karrer and Newman, 2011) supplies one.

The mathematics behind it: the degree-corrected evidence
(this box can be skipped)

Let each node carry a propensity $\theta_i = k_i/\sqrt{2m}$ and let the expected number of links between i and j be $\theta_i\theta_j\,\omega_{g_i g_j}$, where g_i is the block of i. Setting every $\omega_{rs} = 1$ reproduces the configuration model exactly — a network with the same degrees and nothing else — so ω_{rs} measures departure from chance. The expected block-pair weights are

$$\Omega_{rr} = \frac{\kappa_r^2}{4m}, \qquad \Omega_{rs} = \frac{\kappa_r\kappa_s}{2m} \quad (r \neq s). \tag{6.18}$$

Giving each ω_{rs} a Gamma(α, α) prior, of mean one and so centred on the configuration model, and integrating it out,

$$Z(e, \Omega) = \frac{\alpha^\alpha}{\Gamma(\alpha)} \int_0^\infty \omega^{e+\alpha-1} e^{-(\Omega+\alpha)\omega}\, d\omega = \frac{\alpha^\alpha}{\Gamma(\alpha)} \frac{\Gamma(e+\alpha)}{(\Omega+\alpha)^{e+\alpha}}. \tag{6.19}$$

The evidence ratio for the split against no split is then

$$R_{\text{dc}} = \frac{Z(e_{11}, \Omega_{11})\, Z(e_{22}, \Omega_{22})\, Z(e_{12}, \Omega_{12})}{Z(m,\, m)}. \tag{6.20}$$

For large counts, Stirling turns this into something recognizable,

$$\log R_{\text{dc}} \simeq \sum_{r \leq s} e_{rs} \,\ln \frac{e_{rs}}{\Omega_{rs}}, \tag{6.21}$$

the surprise — in the information-theorist's sense — of finding the links where they are rather than where the degrees alone would have put them. This is the Bayesian, degree-corrected counterpart of modularity.

Everything now follows from two ratios. Write $u_{\text{in}} = e_{rr}/\Omega_{rr}$ for the density inside a block relative to chance, and $u_{\times} = e_{12}/\Omega_{12}$ for the seam. Inserting Eq. (6.14) into Eq. (6.18) gives them exactly, as rational functions of $x = 2^t$:

$$u_{\text{in}} = \frac{2x^2}{(x \pm 1)^2}, \qquad u_{\times} = \frac{2x}{x^2 - 1}. \tag{6.22}$$

The mathematics behind it: the evidence per edge *(this box can be skipped)*

Under the flow, Eq. (6.22) converges to the *community fixed point* (Figure 6.5b),

$$(u_{\text{in}}, u_{\times}) \longrightarrow (K, 0), \tag{6.23}$$

with $K = b$ the number of communities, the seam density decaying as $(\lambda_{\text{seam}}/\lambda_{\text{vol}})^t = (1/s)^t$. At the fixed point each within-block term of Eq. (6.21) contributes $\frac{m}{2} \ln K$ and the cross term is subleading, so the evidence density converges to a pure number:

$$\frac{\log R_{\text{dc}}}{m} \longrightarrow \ln K. \tag{6.24}$$

The reason is pure counting, and owes nothing to the diamond. If K equal, well-separated communities hold all the links, each holds $e_{rr} = m/K$ of them, while the configuration model — knowing only the degrees — expects $\Omega_{rr} = m/K^2$. The ratio is K identically, and every link contributes exactly $\ln K$: the entropy of resolving which of K communities it belongs to. This is the same $\ln 2$ per link that broke the ring.

Equation (6.24) explains the emergence of communities in one line, and it is the answer this chapter has been missing. The evidence for a partition is *extensive*: it grows in proportion to the number of links, each one contributing its $\ln K$. The price of positing the partition — the extra block affinities a model must fit — grows only like the *logarithm* of the size. A quantity growing like $m \ln K$ must

eventually overtake one growing like $\ln m$, and once it does it never looks back. Emergence is not a delicate phenomenon requiring a special mechanism; it is arithmetic. All that remains to ask is *where* the crossing happens:

$$r_\kappa(q) = n_t\big|_{t_\star}, \qquad t_\star = \min\Big\{t : \log R_{\rm dc}(t) \geq \ln \tfrac{q}{1-q}\Big\}, \tag{6.25}$$

the threshold $\ln[q/(1-q)]$ being set by the certainty q we demand (Figure 6.5a). With $\log R_{\rm dc} \simeq (\ln K)(bs)^t$ from Eq. (6.24), the crossing generation is

$$t_\star \simeq \left\lceil \log_{bs} \frac{\ln[q/(1-q)]}{\ln K} \right\rceil. \tag{6.26}$$

This is the precise sense in which the Ramsey number is a renormalization-group crossing: it is the generation at which a running, extensive evidence — driven by the volume eigenvalue, once the flow has turned onto the community fixed point — first clears a fixed detection threshold.

The whole (b, s) family can be done at once, and the result is the closest thing this book has to a formula for emergence.

The mathematics behind it: the Ramsey number in closed form *(this box can be skipped)*

The key structural fact is a lemma: for $s \geq 2$, at every generation each link either lies inside a single bundle or joins a pole to a bundle node; no link ever joins two distinct bundles. (Induction on the replacement: a link inside a bundle spawns links inside that bundle, and a pole-to-bundle link spawns pole-to-bundle links and bundle-interior links.) So write c_t for the number of pole-to-bundle links of one bundle and i_t for its interior links. Replacement gives the triangular recursion

$$c_{t+1} = b\, c_t, \qquad i_{t+1} = bs\, i_t + b(s-1)\, c_t, \tag{6.27}$$

whose spectrum can be read straight off the diagonal — $\lambda_{\rm vol} = bs$ and $\lambda_{\rm seam} = b$, now proved for all (b, s), not just the symmetric cell where the two roles cannot be told apart. With $c_1 = 2$, $i_1 = s - 2$,

$$c_t = 2\, b^{\,t-1}, \qquad i_t = b^{\,t-1}\big(s^{\,t} - 2\big). \tag{6.28}$$

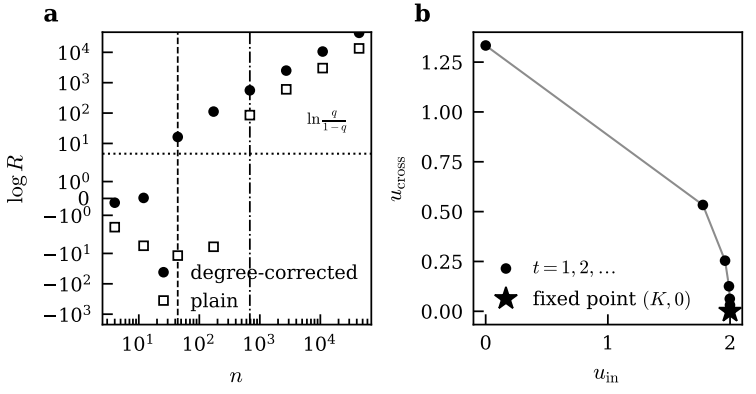


Figure 6.5: **The Ramsey number as a crossing.** (a) Evidence for the split on the diamond lattice against size. r_κ is simply where the curve clears the threshold $\ln[q/(1-q)]$ (dotted, $q = 0.99$): at $n = 44$ for a degree-corrected block model (dashed line) but only at $n = 684$ for a plain one (dash-dotted). (b) The densities of Eq. (6.22) flow to the community fixed point $(K, 0)$: the interiors settle at $K = 2$ times chance, the seam fades to nothing.

The block statistics follow: $e_{00} = i_t + c_t = b^{t-1}s^t$ for the pole block, $e_{rr} = i_t$ and $e_{0r} = c_t$ for each bundle $r \geq 1$, and $e_{rs} = 0$ for distinct bundles. The degree sums are $\kappa_r = 2b^{t-1}(s^t-1)$ and $\kappa_0 = 2b^{t-1}(s^t-1+b)$, and since each replaced link adds $b(s-1)$ nodes,

$$n_t = 2 + \frac{b(s-1)\big[(bs)^t - 1\big]}{bs - 1}. \tag{6.29}$$

The densities, with $x = s^t$, are again rational and again flow to $(b, 0)$:

$$\frac{e_{00}}{\Omega_{00}} = \frac{b\,x^2}{(x+b-1)^2}, \quad \frac{e_{rr}}{\Omega_{rr}} = \frac{b\,x(x-2)}{(x-1)^2}, \quad \frac{e_{0r}}{\Omega_{0r}} \to \frac{b}{s^t}. \tag{6.30}$$

The $\ln b$ contributions sum to $m \ln b$ *exactly*, and the only deficit is the seam's — the price of the b^t crossing links still present:

$$\log R_{\text{dc}} = m \ln b \,\big[1 - \delta_t\big], \qquad \delta_t = \frac{2(b-1)}{b \ln b}\,\frac{t \ln s + 1}{s^t}. \tag{6.31}$$

Setting this equal to the threshold and inverting Eq. (6.29),

$$r_\kappa(b,s;q) = 2+\frac{b(s-1)}{bs-1}\Big[(bs)^{t_\star}-1\Big], \qquad t_\star = \Big\lceil \log_{bs}\Big(\ln\tfrac{q}{1-q}\Big/\ln b\Big)\Big\rceil. \tag{6.32}$$

Equation (6.32) deserves a moment. The Ramsey community number, which three sections ago was a number we could only obtain by simulating thousands of networks, is here a formula. And look at what survives in it: b, through the entropy accumulated per link, and s, through the volume growth. The degrees drop out. The priors drop out. The details of the lattice drop out. Emergence, it turns out, depends on almost nothing about the network except how fast its communities grow and how fast their boundary does.

6.9 Communities, or just degrees?

One doubt remains, and it is the sharpest of all. Every network in this book is hub-dominated. A block model handed a scale-free network can improve its fit simply by putting the hubs in one group and the rest in another — not because there is any community structure, but because there is a degree distribution to absorb. How do we know that what we have been calling communities is not just the hubs in disguise?

The pseudofractal web settles it. Its rule is as local as they come: start from a triangle, and at every generation let each existing link spawn a new node joined to both of its ends, keeping all the old links. What grows is deterministic, and elementary bookkeeping gives

$$n_t = \tfrac{3}{2}\big(3^t+1\big), \qquad E_t = 3^{\,t+1}, \tag{6.33}$$

so the mean degree tends to four. A node born at generation τ ends with degree $2^{\,t-\tau+1}$, and counting nodes by birth generation gives a power-law degree distribution, $p(k)\sim k^{-\gamma}$ with

$$\gamma = 1+\frac{\ln 3}{\ln 2} \simeq 2.585, \tag{6.34}$$

squarely in the scale-free range of real networks. Three seed hubs of

degree 2^{t+1} tower over everything. This web has exactly the degree heterogeneity that makes the doubt bite.

It also has two rival splits with completely different characters. The *branch* cut follows the recursive construction, separating one branch of the web from the others. The *hub–leaf* cut ignores the construction entirely and simply puts the high-degree nodes on one side.

The mathematics behind it: the branch cut, and what a degree-blind model makes of it *(this box can be skipped)*

Label the three seed links $0, 1, 2$; every new node and both its links inherit the label of the link that spawned it, so each non-seed node sits in exactly one branch. Take block 1 to be one branch plus the three seed hubs, block 2 the other two branches. The counts are exact:

$$\begin{aligned} n_1 &= \tfrac{1}{2}\big(3^t + 5\big), & n_2 &= 3^t - 1, \\ E_1 &= 3^t + 2, & E_2 &= 2\cdot 3^t + 2 - 2^{\,t+2}, \\ E_{12} &= 2^{\,t+2} - 4, \end{aligned} \tag{6.35}$$

with $n_1 + n_2 = n_t$ and $E_1 + E_2 + E_{12} = E_t$. The seam is thin: E_{12} grows only as $2^t = n^{\ln 2/\ln 3} \simeq n^{0.631}$, so the fraction of crossing links $E_{12}/E_t \sim (2/3)^t \to 0$ — the structural fact that drives everything. Feeding Eq. (6.35) into the plain evidence Eq. (6.3) and applying Stirling gives a linear growth,

$$\log R \sim \Big(\ln 3 - \tfrac{2}{3}\ln 2\Big)\, n \approx 0.6365\, n, \tag{6.36}$$

independent of the prior.

Now correct for the degrees. The block degree sums of the branch cut are

$$\kappa_1 = 2\cdot 3^t + 2^{\,t+2}, \qquad \kappa_2 = 4\cdot 3^t - 2^{\,t+2}, \tag{6.37}$$

with $\kappa_1 + \kappa_2 = 2E_t$, and inserting these into Eq. (6.18) gives densities that *exceed* the configuration model's expectation inside the blocks while the crossing density vanishes:

$$\frac{E_1}{\Omega_{11}} \to 3, \qquad \frac{E_2}{\Omega_{22}} \to \tfrac{3}{2}, \qquad \frac{E_{12}}{\Omega_{12}} \sim \Big(\tfrac{2}{3}\Big)^t \to 0. \tag{6.38}$$

The branch cut therefore survives the correction, and Eq. (6.21)

gives its slope:

$$\log R_{\mathrm{dc}} \sim \left(2\ln 3 - \tfrac{4}{3}\ln 2\right) n \approx 1.2730\, n, \tag{6.39}$$

exactly *twice* the plain slope of Eq. (6.36). Correcting for the degrees does not weaken the case for these communities. It doubles it.

The result is a reversal, and it is damning for the naive method. The plain block model prefers the hub–leaf cut — it is chasing the degree gradient, exactly as feared. Correct for degrees and the hub–leaf cut collapses while the branch cut strengthens: the recursive communities are real, and they are what a degree-blind detector was missing while it stared at the hubs. The two verdicts on *when* disagree just as violently (Figure 6.6). The plain model reports no communities until $n = 1095$; the corrected model has been sure of them since $n = 42$. Same network, same partition, same question — and a twenty-five-fold disagreement about the answer. The direction of that disagreement is the counterintuitive part: the degree gradient, far from manufacturing the split, *delays* the detector that cannot see past it.

The moral is not that one number is right and the other wrong. It is that r_κ was never a property of the network alone. It is a property of the network together with the question we ask of it, and the warning lodged when we defined it now has to be paid. The exactly solvable cases are valuable precisely because they let us see this: a quantity that looked like a fact about the web turns out to depend, by a factor of twenty-five, on which null model we hold it against. That does not dissolve the chapter's thesis — under *either* rule the local construction has a finite r_κ, while a network with no local structure has none — but it does mean that a bare Ramsey number, quoted without its detection rule, says less than it appears to.

6.10 How many communities?

We have been asking whether a network splits in two. That was a convenience, and it is time to drop it. A self-similar network has structure at every scale, so why stop at the first cut?

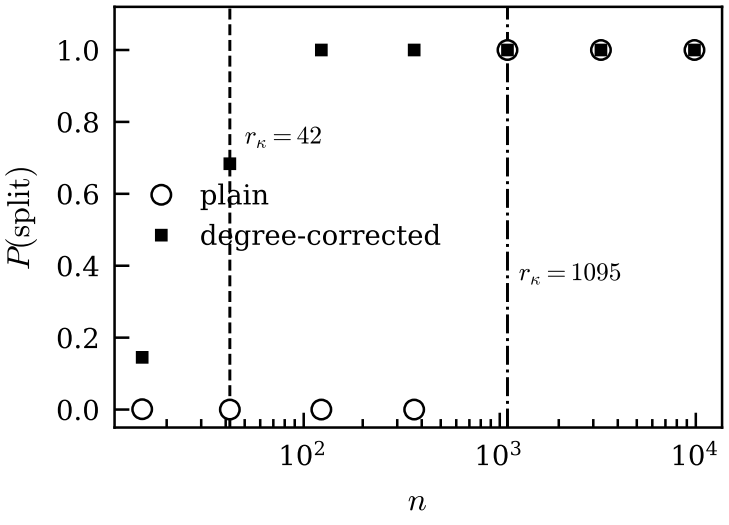


Figure 6.6: **Communities, or just degrees?** The pseudofractal web under a plain block model (open circles) and one corrected for the degree sequence (filled squares). The corrected model is certain of the recursive communities from $n = 42$; the plain one denies them until $n = 1095$. The degree gradient, far from manufacturing the split, *delays* the detector that cannot see past it.

Take the pseudofractal web again, and cut it three ways instead of two, one block per branch. The symmetry now works for us rather than against us, and the counts are cleaner than the branch cut's:

$$n_r = \tfrac{1}{2}\big(3^t + 1\big), \quad \kappa_r = 2 \cdot 3^t, \quad e_{rr} = 3^t - 2^t, \quad e_{rs} = 2^t. \tag{6.40}$$

The densities behave exactly as the diamond's fixed point promised, with $K = 3$:

$$\frac{e_{rr}}{\Omega_{rr}} = 3\big(1 - (2/3)^t\big) \to 3, \qquad \frac{e_{rs}}{\Omega_{rs}} = \tfrac{3}{2}(2/3)^t \to 0, \tag{6.41}$$

and the evidence density is $\ln 3$ per link, just as Eq. (6.24) requires

of three well-separated communities:

$$\log R_{\mathrm{dc}}^{(3)} \sim 2\ln 3\, n \approx 2.197\, n. \tag{6.42}$$

This beats the two-block cut of Eq. (6.39) by exactly $\frac{4}{3}\ln 2\, n$ — the evidence the branch cut forfeited by lumping two communities together. Three blocks describe the web better than two. And having seen that, there is no reason to stop: removing the links of generation d leaves 3^{d+1} statistically identical sub-webs, and each level of refinement adds another $\simeq 2\ln 3\, n$ to the evidence.

What stops it? The seams do. Every cut has to pay for the links it severs, and the deeper the cut the more of them there are. On the diamond the competition can be written down exactly. A depth-d cut severs $c(d) = 2^{t+d}$ links, leaving a fraction $e_{\mathrm{in}}/m = 1 - 2^{d-t}$ inside blocks — each level doubling the interface, which is the seam eigenvalue $\lambda_{\mathrm{seam}} = b = 2$ acting once per level. Balancing that against the reward for splitting the degree sum evenly gives a modularity

$$Q(d) = \frac{e_{\mathrm{in}}}{m} - \gamma\frac{\sum_c \kappa_c^2}{(2m)^2} = 1 - 2^{\,d-t} - \gamma\, 2^{-d} + O(2^{-t}), \tag{6.43}$$

where γ is the resolution at which we choose to look. Maximizing over the depth gives $2^{2d} = \gamma 2^t$, that is

$$d_\star = \tfrac{1}{2}\big(t + \log_2\gamma\big), \qquad q_{\mathrm{opt}} = 2^{d_\star} = \sqrt{\gamma}\; 2^{t/2} \sim \sqrt{\gamma\, n}. \tag{6.44}$$

The number of communities that best describes the network is not two, and is not any fixed number: it *grows with the network*, as its square root. The web does the same, with $K_{\mathrm{opt}} = 3^{\lfloor t/2\rfloor} \sim \sqrt{n}$. The optimum sits at the geometric midpoint of the network's own hierarchy — about $\sqrt{n}$ communities of about $\sqrt{n}$ nodes each — and the two-block cut we laboured over is merely the top of that tree. It is worth noting that $\sqrt{n}$ is the same scale that Chapter 7 will find in the exponent of the shortest path multiplicity on this very lattice; both are the seam scale b^t in disguise.

6.11 Communities as a state of matter

There is one more thing the diamond will give us, and it is the strangest. So far communities have been an inference: the parti-

tion a statistician prefers. But the quantity being optimized is a Hamiltonian, and a Hamiltonian is physics. Written in the form of Reichardt and Bornholdt (2004),

$$\mathcal{H} = -\sum_{i\neq j} \left(A_{ij} - \gamma\, p_{ij}\right) \delta(\sigma_i, \sigma_j), \qquad p_{ij} = \frac{\kappa_i \kappa_j}{2m}, \tag{6.45}$$

where σ_i labels the community of node i and A_{ij} is the adjacency matrix, the ground state is the community partition, and at $\gamma = 1$ this is modularity itself. For two communities the labels are Ising spins, and the Hamiltonian rearranges into something a physicist recognizes at sight:

$$\mathcal{H} = -\sum_{\langle ij\rangle} \sigma_i \sigma_j + \frac{\gamma}{4m}\Big[M_\kappa^2 - \sum_i \kappa_i^2\Big], \qquad M_\kappa = \sum_i \kappa_i \sigma_i. \tag{6.46}$$

A ferromagnet — neighbors want to agree, which is cohesion — plus an infinite-range, degree-weighted penalty that punishes the whole network for agreeing too much, which is what forces a split.

Because the penalty couples the spins only through the single collective variable M_κ, the model can be solved exactly on the diamond by decimation, and the answer is a genuine phase. Take the correlation between the two poles, $-\langle \sigma_A \sigma_B \rangle$, as the order parameter. Below the ferromagnetic critical temperature

$$\frac{k_B T_c}{J} = 1.641, \tag{6.47}$$

the two poles lock into *opposite* communities for any resolution $\gamma > 0$, however small — the degree-weighted penalty bites hardest on the two highest-degree nodes — and the order survives as $n \to \infty$. The resolution selects *which* partition orders, not *whether* it does. The contrast with a plain ferromagnet is the whole point: an ordinary Ising model on the same lattice has no such phase, its ground state being the trivial single community. It is the community penalty that changes the ground state into a partition.

Push to the growing hierarchy of the last section and the picture completes. With $q = 2^d$ colors the model becomes a Potts model,

whose series and parallel decimations are

$$\begin{aligned} A_{\text{ser}} &= A_1A_2 + (q{-}1)B_1B_2, \\ B_{\text{ser}} &= A_1B_2 + B_1A_2 + (q{-}2)B_1B_2, \\ A_{\text{par}} &= A_1A_2, \qquad B_{\text{par}} = B_1B_2, \end{aligned} \tag{6.48}$$

with A the weight for two ends in the same community and B for different ones. Solving these, every $q > 2$ orders through a *first-order* transition — the partition snaps into place rather than fading in, as is known for Potts models on hierarchical lattices (Griffiths and Kaufman, 1982) — at a temperature that falls only logarithmically with the number of communities,

$$\frac{k_B T_c(q)}{J} \sim \frac{1}{c\,\ln q}, \qquad c \approx 0.70. \tag{6.49}$$

So the hierarchy orders level by level, coarsest first, each finer level freezing at a lower temperature; and since $q_{\text{opt}} \sim \sqrt{n}$, the finest level still orders at $k_B T_c/J \sim 1/\ln n$. Every thermodynamically stable level of the hierarchy survives as the network grows.

The conclusion is worth stating baldly. On this lattice the community structure is not merely detectable, in the sense that a statistician would infer it. It is *ordered*, in the sense that a physicist would recognize: a phase of matter, with a critical temperature, an order parameter, and a cascade of transitions building the hierarchy from the top down. Communities, born of nothing but local wiring, turn out to be a state that a network can be in.

6.12 Summary

Community structure, long blamed on built-in differences between nodes, is instead something that *emerges* from growing a network by local rules, even when every node is identical. Made precise through the Ramsey community number r_κ — the size beyond which communities are all but certain — the statement is clean: the local rules of earlier chapters (search and duplication–split) have a finite r_κ, while their degree-preserving shuffles, and the non-local Barabási–Albert model, have none. Triangles are not required (duplication–split makes none and still segregates), but locality is:

communities come soonest for rules that close short cycles between nearby nodes, and disappear once links may span the whole network.

Three networks simple enough to solve exactly turn all of that from a measurement into a theory. A featureless ring of beads, with no growth and no randomness whatever, is already split in two once each bead reaches beyond its nearest neighbor. The evidence for the split is $\log R \simeq (m_{\mathrm{e}} - n)\ln 2$: a gain of $\ln 2$ for every link, against a price of $n \ln 2$ for saying which node goes where. So a network is broken into communities precisely when it carries more links than nodes — which is to say, when it carries an extensive number of independent cycles. A tree never breaks; a single cycle sits exactly on the fence, which is why the plain cycle is not reluctant to split but exactly balanced. The same short loops that Chapter 7 shows multiplying shortest paths are what carve a network into communities.

The diamond lattice says why any threshold exists at all. Its block-model tallies renormalize exactly, under a map with just two eigenvalues — bs for the bulk of a community, b for the seam between them — whose ratio $1/s$ decides everything: for $s \geq 2$ the seam is left behind and the communities sharpen without limit, while for $s = 1$ it keeps pace and the lattice never separates, the exact counterpart of the plain cycle. The densities flow to a fixed point where each community is K times denser than chance and the seam has vanished, so every link contributes $\ln K$ — the entropy of resolving which community it belongs to — and the evidence is *extensive* while the price of positing a partition grows only *logarithmically*. A quantity growing like $m \ln K$ must overtake one growing like $\ln m$. Emergence is not delicate; it is arithmetic, and r_κ merely records where the crossing falls — now a closed formula in which the degrees, the priors, and the lattice all drop out, leaving only how fast the communities grow and how fast their boundary does.

The pseudofractal web supplies the caution and the reassurance together. Correct for the degree sequence — score the split against a null that already knows every node's degree — and the same web's Ramsey number moves from 1095 to 42, while the preferred cut flips from the hubs to the branches. The communities are real: they not only survive the correction but double their evidence under it, and the degree gradient turns out to *delay* a degree-blind detector

rather than fool it. But r_κ belongs to the network and the question together, never to the network alone.

Two closing surprises. Asking how many communities, rather than whether there are two, gives an answer that grows with the network: about $\sqrt{n}$ communities of about $\sqrt{n}$ nodes, the geometric midpoint of the network's own hierarchy, and the same $\sqrt{n}$ that governs shortest path multiplicity on the same lattice. And the partition is not merely something a statistician infers: written as a Hamiltonian it has a critical temperature, an order parameter, and a cascade of first-order transitions that build the hierarchy level by level. Communities are a state the network is in.

Having already accounted for degree distributions, the clustering hierarchy, and degree correlations, this chapter adds community structure to the list of global patterns that locality alone can explain (Vazquez, 2002).

Chapter 7

Boosting shortest path multiplicity

7.1 The redundancy of shortest paths

The small-world property, met back in the introduction, promises that any two nodes of a real network are joined by a *short* path — only a few hops apart. It says nothing, though, about how *many* short paths there are. That count turns out to matter a great deal. If the only shortest route between two nodes runs through a single chain of intermediaries, the connection is fragile: cut one link and the shortest route is gone. If instead there are several shortest routes side by side, the connection is robust, and whatever flows between the two nodes — a message, a signal, a delay — can travel along all of them at once. The quantity that captures this is the *shortest path multiplicity*: the number of distinct shortest routes between a pair of nodes. A simple picture: if two friends can each be reached from you in two handshakes, but by two completely different chains of acquaintances, the multiplicity of that connection is two. In real networks this multiplicity is conspicuously high.

And it is not an isolated fact. Deng et al. (2026) found that, across many real networks, the abundance of multiple shortest paths is strongly correlated with *community structure* — the very groupings of the previous chapter — more strongly than with any other

standard measure. Two networks of the same size and the same degree distribution can differ widely in their path multiplicity, and the difference tracks how community-rich they are. A correlation like that begs for an explanation. Does community structure somehow *cause* paths to multiply, or do both spring from a common deeper source? The previous chapter hands us the obvious suspect: local growth rules, which we saw conjure communities out of nothing but locality. This chapter, following Vazquez (2026a), completes the case by showing that those same local rules also *boost* the multiplicity of shortest paths — so that the striking link between paths and communities is not one causing the other, but the shared shadow cast by the rule that grew the network.

The mechanism is easy to picture once named. Local rules build short *cycles* — closed loops — as a matter of course: a triangle whenever a friend introduces a friend, a square whenever a gene is copied. And a cycle is exactly a structure that offers more than one way around. In a network with no loops at all — a *tree* — there is precisely one route between any two nodes; the more loops a network carries, the more alternative shortest routes it affords. Path multiplicity is simply what local rules leave behind when they close loops.

7.2 Defining the multiplicity

We need only one quantity. Write $\mu_{ij}(G)$ for the number of distinct shortest paths between nodes i and j in a network G — counting every route that ties for the minimum number of hops. Averaging this over all pairs of nodes, and over many networks grown the same way, gives the *average shortest path multiplicity* $\langle \mu \rangle$, the single number we will follow as the network grows to size n. A tree, with its unique routes, has $\langle \mu \rangle = 1$ exactly; anything above one measures the redundancy that loops provide. The whole chapter is about one question: how fast does $\langle \mu \rangle$ rise as the network gets bigger, and does that rate betray whether the network was grown by a local rule?

Four growth rules will do all the work, and we have met them all before. Their definitions matter here more than they have so far, because we are about to compute their exponents, so let us state them exactly.

Local search, $LS(n, m)$, the search rule of Chapter 2 in undirected form. Start from two connected nodes. To add a node, pick an existing node at random, take an m-step random walk from it, and link the newcomer to every node the walk visited. It carries preferential attachment because the chance of being visited, beyond the entry node, is proportional to degree; and it makes at least one triangle at every step, between the entry point and the first neighbor visited. For $m = 1$ it *is* the triadic-closure rule of Chapter 3.

Duplication–split, $DS(n, q)$, the undirected cousin of the project rule of Chapter 5. Start from a cycle of four nodes, so that no node has degree below two and no cycle is shorter than four. To add a node i, pick an existing node j at random. With probability q, make i a duplicate of j, linking i to every neighbor of j. Otherwise pick a random neighbor k of j and *split* the link: delete (j, k) and create (i, j) and (i, k). It carries preferential attachment because the chance that a node has a neighbor duplicated is proportional to its degree. Duplication makes no triangles and splitting destroys any that might form; its motif is the square.

The *bubble* model, $BB(n, L)$. Start from a cycle of $L + 2$ nodes. At each step add a chain of L new nodes and attach the two ends of the chain to the two ends of a randomly chosen existing link, closing a fresh cycle of length $L+2$. It carries preferential attachment because the chance of sitting at the end of the chosen link is proportional to degree. $BB(n, 1)$ is the Dorogovtsev–Mendes rule.

Against these we set the *Barabási–Albert* model $BA(n, m)$: start from a complete graph of $m+1$ nodes, and let every newcomer attach m links to nodes chosen from the *whole* network with probability proportional to degree. It is our non-local control.

7.3 The baseline: random networks

To judge what counts as "boosted," we first need the expectation in the *absence* of any local structure. The natural reference is a random network, and here we can do better than simulate — we can calculate. The answer is remarkably simple, and it turns on a

single property of the degree distribution: the *average excess degree*

$$c_2 = \frac{\langle k(k-1)\rangle}{\langle k\rangle}, \tag{7.1}$$

which is the average number of *further* links you find when you arrive at a node by following one of its links — its degree, less the link you came in on. It is the natural branching factor of a network: from where you stand, c_2 is how many ways the path onward can continue.

The cleanest way to see where Eq. (7.1) comes from is to do the simplest case first — the classic random graph of Erdős and Rényi, in which every one of the $\binom{n}{2}$ possible links is present with the same probability p — and then repair it for a general degree distribution.

The mathematics behind it: counting the shortest paths of a random graph *(this box can be skipped)*

Write $c = np$ for the mean degree and $D = \ln n / \ln c$ for the typical distance between two nodes, and assume $c > 1$ so the graph is connected. Fix two nodes. A path of length d between them uses $d-1$ intermediate nodes, in order, drawn from the remaining $n-2$, so the number of *candidate* paths is

$$m_d = (n-2)(n-3)\cdots(n-d) \approx n^{\,d-1} \tag{7.2}$$

for $d \ll n$. Each candidate is *realized* only if all d of its links happen to be present, which has probability p^d; treating the candidates as independent, the number realized is binomial, with mean

$$\langle M_d\rangle = m_d\, p^{\,d} \approx \frac{(np)^d}{n} = c^{\,d-D}. \tag{7.3}$$

The shortest paths are those at the smallest d for which any path exists at all, so we must weight $\langle M_d\rangle$ by the probability F_{d-1} that nothing shorter exists:

$$\langle \mu\rangle = \sum_d \langle M_d\rangle\, F_{d-1}. \tag{7.4}$$

Since $\Pr(M_d = 0) = (1-p^d)^{m_d} \approx e^{-\langle M_d\rangle}$ for $p \ll 1$,

$$F_d = \prod_{k=1}^{d} \Pr(M_k = 0) = \exp\Big(-\sum_{k=1}^{d}\langle M_k\rangle\Big). \tag{7.5}$$

The sum in the exponent is a geometric series of ratio $c > 1$, so it is dominated by its last term, $\sum_{k \le d} \langle M_k \rangle \approx c^{\,d-D}$, and

$$F_d \approx \exp\big(-c^{\,d-D}\big). \tag{7.6}$$

Substituting Eqs. (7.3) and (7.6) into Eq. (7.4),

$$\langle \mu \rangle \approx \sum_d c^{\,d-D} \exp\big(-c^{\,d-D-1}\big). \tag{7.7}$$

The summand is the function $xe^{-x/c}$ evaluated along a geometric ladder: it climbs by a factor c per hop while $d < D$, and is then cut down super-exponentially. It therefore peaks sharply, one hop beyond the typical distance, at $d = D + 1$, and evaluating it there leaves

$$\langle \mu \rangle \approx \frac{c}{e} + O(1), \tag{7.8}$$

the e being simply the value of xe^{-x} at its maximum.

Equation (7.8) already tells us three things. If the graph is *sparse* — c fixed as n grows, which is the case for every real network — the multiplicity is a *constant*. If it is *minimally connected*, $c = b \ln n$, the multiplicity grows as $(b/e) \ln n$. And if it is *dense*, $c \sim n$, it grows in proportion to n, recovering a result of Dong et al. (2025). But a real network is sparse *and* hub-dominated, and an Erdős–Rényi graph is neither. We need the same calculation for an arbitrary degree distribution.

The mathematics behind it: the same count for any degree distribution *(this box can be skipped)*

Let the degrees follow π_k, with mean $c_1 = \sum_k \pi_k k$ and excess degree $c_2 = \sum_k \pi_k k(k-1)/c_1$. (We write π_k rather than p_k here because p is already the probability that a link exists.) The calculation runs exactly as before; only the probability that the next link exists changes.

Build a candidate path node by node. The first step is from a node picked at random to another node picked at random, which are linked with probability $p_1 = c_1/n$, with associated distance $D_1 = \ln n / \ln c_1$. Every step after the first is different, because we now stand at the

end of a link, and a node reached by following a link is not a typical node: its degree is distributed as $k\pi_k/c_1$, and one of those links is the one we arrived on, leaving $k-1$. The next node is picked at random and has degree k' with probability $\pi_{k'}$; the chance the two are linked is $kk'/(nc_1)$, which on averaging over both gives

$$p_2 = \frac{c_2 c_1}{n c_1} = \frac{c_2}{n}, \qquad D_2 = \frac{\ln n}{\ln c_2}. \tag{7.9}$$

So a path of length d costs one factor of p_1 and $d-1$ factors of p_2, and Eqs. (7.3) and (7.6) become

$$\langle M_d \rangle \approx c_1 c_2^{d-D_2-1}, \qquad F_d \approx \exp\big(-c_1 c_2^{d-D_2-1}\big), \tag{7.10}$$

for $d > 1$. Substituting into Eq. (7.4),

$$\langle \mu \rangle \approx \sum_d c_1 c_2^{d-D_2-1} \exp\big(-c_1 c_2^{d-D_2-2}\big), \tag{7.11}$$

whose summand again peaks sharply, now where $c_1 c_2^{d-D_2-2} = 1$, leaving

$$\langle \mu \rangle \approx \frac{c_2}{e} + O(1). \tag{7.12}$$

For an Erdős–Rényi graph $c_2 = c_1 = c$ and we recover Eq. (7.8).

Equation (7.12) is the baseline, and it is worth pausing on how much it explains. The multiplicity of a random network is simply its branching factor, divided by e. The mean degree does not appear; the whole distribution of degrees enters through the single number c_2. And notice what happened between the two boxes: c_1 was replaced by c_2 everywhere it mattered, because after the first step a path is always standing at the end of a link, where the branching is the *excess* degree and not the degree.

Everything now hinges on how c_2 behaves as the network grows, and since c_2 involves $\langle k^2 \rangle$, that is a question about hubs.

The mathematics behind it: how the excess degree grows with size *(this box can be skipped)*

Take a scale-free network, $p(k) \sim k^{-\gamma}$. For $\gamma > 3$ the second moment $\langle k^2 \rangle$ converges, so c_2 tends to a constant and there is nothing to

discuss. For $\gamma \leq 3$ it diverges, and the divergence is cut off only by the largest degree actually present. That largest degree follows from asking which k is rare enough to appear about once in a network of n nodes, $p(k_{\max}) = 1/n$:

$$k_{\max} \sim n^{1/\gamma}. \tag{7.13}$$

Now do the sum. For $\gamma = 3$ the summand $k^2 p(k) \sim 1/k$, so

$$c_2 \sim \sum_{k=1}^{k_{\max}} \frac{1}{k} \sim \ln k_{\max} \sim \ln n, \tag{7.14}$$

the logarithm of the size. For $2 < \gamma < 3$ the sum is dominated by its upper end, $c_2 \sim k_{\max}^{3-\gamma}$, and using Eq. (7.13),

$$c_2 \sim n^{-1+3/\gamma}. \tag{7.15}$$

It is worth doing the $\gamma = 3$ case explicitly, since three of our models share it. There $p_k = A\,\Gamma(k)/\Gamma(3+k) = A/[(2+k)(1+k)k]$, so

$$\langle k \rangle \sim A \sum_{k\geq 2} \frac{1}{(2+k)(1+k)} = A \sum_{k\geq 2} \Big(\frac{1}{k+1} - \frac{1}{k+2} \Big) \sim \frac{A}{3}, \tag{7.16}$$

a telescoping sum that converges, while

$$\langle k^2 \rangle \sim A \sum_{k=1}^{k_{\max}} \frac{1}{k} \sim A \ln k_{\max} \sim \frac{A}{3} \ln n \tag{7.17}$$

does not. Their ratio is $c_2 \sim \ln n$, confirming Eq. (7.14).

Feeding the three cases into Eq. (7.12) gives the baseline in full:

$$\langle \mu \rangle = a, \qquad \gamma > 3, \tag{7.18}$$

$$\langle \mu \rangle = a + \tfrac{1}{e} \ln n, \qquad \gamma = 3, \tag{7.19}$$

$$\langle \mu \rangle = a + b\, n^{\alpha}, \qquad \alpha = \tfrac{3}{\gamma} - 1, \qquad 2 < \gamma < 3. \tag{7.20}$$

When $\gamma > 3$ the multiplicity *stops growing altogether*. At $\gamma = 3$ it creeps up like $\ln n$ — about as slow as growth gets, since to add a fixed amount you must *multiply* the size — and the predicted slope is not merely logarithmic but exactly $1/e$. Below $\gamma = 3$ the hubs are heavy enough to drive a genuine power law. So there is no single

random baseline: the bar that a local rule must clear is set by the degree distribution it happens to produce, and we must compare like with like.

7.4 Which rule makes which exponent

That last sentence obliges us to know the exponent of every model in this chapter. Happily one calculation settles them all at once, because all four rules — and the shuffled versions we shall compare them against — attach in the same way: a node gains links at a rate that is affine in its degree.

The mathematics behind it: one rate equation for every model *(this box can be skipped)*

Let $n_k(n)$ be the expected number of nodes of degree k once the network has grown to n nodes. Suppose a node of degree k gains a link at rate $(a+bk)/n$ — a constant part a and a preferential part b — and that the newcomer arrives either with degree 2, with probability c, or as a duplicate of an existing node, at rate d/n. Then

$$n_k(n+1) = n_k(n) + \frac{a+b(k-1)}{n}\, n_{k-1} - \frac{a+bk}{n}\, n_k + c\,\delta_{k2} + \frac{d}{n}\, n_k. \tag{7.21}$$

The first two terms are the traffic in and out of degree k; the last two account for the newcomer's own degree. This is mean field: the expectation after the step is computed from the expectations before it, ignoring the fluctuations of a single addition. Look for a steady state in which the shape of the distribution is fixed and only its size grows, $n_k(n) = n\,p_k$. Substituting and cancelling one power of n gives a recursion in k alone,

$$p_k = \frac{\eta + k - 1}{\nu + k}\, p_{k-1} + \frac{c}{b}\,\frac{1}{\nu+k}\,\delta_{k2}, \qquad \eta = \frac{a}{b}, \quad \nu = \frac{1+a-d}{b}, \tag{7.22}$$

which unrolls into a ratio of Gamma functions,

$$p_k = \frac{c}{b}\,\frac{\Gamma(2+\nu)}{\Gamma(2+\eta)}\,\frac{\Gamma(k+\eta)}{\Gamma(k+1+\nu)}. \tag{7.23}$$

(Check it by induction: it holds at $k = 2$, and substituting p_{k-1} from Eq. (7.23) into Eq. (7.22) returns p_k.) For $k \gg \max(\eta, \nu)$ the ratio of Gammas behaves as $k^{\eta-\nu-1}$, a power law $p_k \sim k^{-\gamma}$ with

$$\gamma = 1 + \frac{1-d}{b}. \tag{7.24}$$

Only two of the four constants survive: the preferential rate b, and the duplication rate d.

Equation (7.24) is a small master key. Read off each rule's constants and its exponent falls out.

For $BA(n, 2)$ every newcomer brings two links and hands them out in proportion to degree, so $a = 0$, $b = 1/2$, $c = 1$, $d = 0$, and

$$\gamma_{BA} = 3. \tag{7.25}$$

For $LS(n, m)$ the walk lands on the entry node uniformly, which is the constant part $a = 1$, and thereafter in proportion to degree, giving $b = m/\langle k \rangle = m/[2(m+1)]$ with $c = 1$, $d = 0$, so that

$$\gamma_{LS} = \frac{3m+2}{m}. \tag{7.26}$$

For $DS(n, q)$ duplication is the only source of preference, $b = q$, and it is also the only way the newcomer copies a degree, $d = q$, with $a = 0$ and $c = 1 - q$; then $\eta = 0$, $\nu = -1 + 1/q$ and

$$\gamma_{DS} = \frac{1}{q}. \tag{7.27}$$

For $BB(n, L)$ the chain's ends land on a random link, so $b = 2/\langle k \rangle = 1/(L+1)$ with $a = 0$, $c = 1$, $d = 0$, $\eta = 0$, $\nu = L + 1$, and

$$\gamma_{BB} = L + 2. \tag{7.28}$$

Two coincidences in this list are worth having. First, $DS(n, q)$ and $BB(n, L)$ produce the *same* degree distribution whenever $q = (L+2)^{-1}$ — so any difference in their multiplicity cannot be blamed on their degrees. Second, and this is what makes the comparisons of the next section fair, $DS(n, 1/3)$, $BB(n, 1)$ and $BA(n, 2)$ all sit at

$\gamma = 3$: three rules, one degree distribution, and — as we are about to see — three quite different multiplicities.

And now the baseline can be made concrete. Take a network grown by a local rule and shuffle its links at random while leaving every node's degree untouched — the same configuration-model scramble used as a control in the previous chapter. The result keeps the degree distribution and destroys everything else; we mark it with a star. By Eq. (7.26) the randomized local-search network $LS(n, 1)^*$ has $\gamma = 5$, and so by Eq. (7.18) its multiplicity must *stop growing*; it duly flattens out (Figure 7.1a). By Eq. (7.27) the randomized $DS(n, 1/3)^*$ has $\gamma = 3$, and creeps up logarithmically, close to the predicted slope $1/e$ (Figure 7.1b). The randomized $DS(n, 2/5)^*$ has $\gamma = 5/2$, for which Eq. (7.20) predicts $\alpha = 3/\gamma - 1 = 1/5$, and it climbs as $n^{1/5}$ (Figure 7.1c). Plotted against c_2 rather than n, all of them — and the Barabási–Albert model with them — collapse onto the single mean-field line c_2/e of Eq. (7.12) (Figure 7.1d). Three regimes, one formula, and no fitted parameter anywhere except the additive constants.

The Barabási–Albert model deserves its own note, because it is a *growing* network yet lands squarely on the random baseline. With $m = 2$ it has $\gamma = 3$ by Eq. (7.25), and its multiplicity creeps up logarithmically, exactly as its reshuffled cousins do. It has no local structure to lose: its newcomers attach across the whole graph and never consult a neighborhood. This echoes the previous chapter, where the very same model failed to develop communities. Lacking locality, it builds neither the modules nor the redundant paths, and on both counts it behaves as though its links had been shuffled at random.

7.5 Local rules boost the multiplicity

Now run the same models *without* shuffling, and compare each against its own starred baseline (Figure 7.2). The multiplicity climbs decisively faster. It helps to keep a ladder of growth rates in mind: constant, then logarithmic, then *log-quadratic*, then *exponential* — each rung leaving the one below it far behind.

Most local rules — local search and the bubble model among

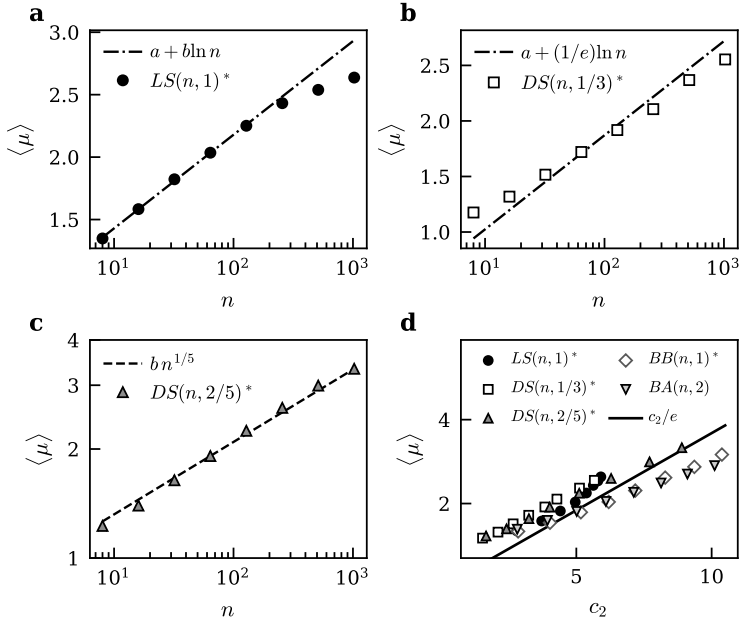


Figure 7.1: **The random baseline.** Average shortest path multiplicity $\langle \mu \rangle$ against network size n for randomized networks (starred), which keep each node's degree but destroy all local structure. (a) $LS(n,1)^*$, degree exponent $\gamma = 5$: growth stalls, the points falling away from the fitted logarithm. (b) $DS(n,1/3)^*$, $\gamma = 3$: logarithmic, near the predicted slope $1/e$. (c) $DS(n,2/5)^*$, $\gamma = 5/2$: the predicted power law $n^{1/5}$. (d) All models, including the non-local $BA(n,2)$, plotted against the excess degree c_2 collapse onto the mean-field line c_2/e of Eq. (7.12).

them — sit on the log-quadratic rung,

$$\langle \mu \rangle = a + b \ln n + c(\ln n)^2, \tag{7.29}$$

the extra squared term bending the curve clearly upward. For $LS(n,1)$ this is a particularly sharp contrast: the unshuffled network grows without limit while the shuffled one, with the very same

degrees, stops growing altogether. Whatever is doing the work is not in the degree distribution at all.

The duplication–split rule is different, and dramatically so: its multiplicity grows *exponentially* with size,

$$\langle \mu \rangle = a\, e^{bn}, \qquad a, b > 0. \tag{7.30}$$

Exponential growth is a different world from logarithmic. Over the same range of sizes in which its shuffled counterpart moves from $\langle \mu \rangle \approx 1.2$ to 2.6, the unshuffled $DS(n, 1/3)$ runs from 1.3 to more than 10^5: by a thousand nodes, a typical pair of nodes is joined by hundreds of thousands of equally short routes. Copying, it turns out, is the great multiplier of shortest paths, and that is what the title of this chapter names.

7.6 Cycles and the boost

Why should duplication outrun the rest? The answer is in the kind of loop each rule makes. The triadic-closure rule of the friends chapter closes *triangles*, loops of length three — but in a triangle the two nodes flanking a third are themselves directly linked, so there is still only *one* shortest route between any pair. A triangle adds cohesion but no extra routes. Duplication, by contrast, closes *squares*, loops of length four, and a square is the smallest loop that genuinely offers a choice: its two opposite corners are two hops apart by *two* different paths, one around each side. That single observation is the seed of the boost; the box pushes it a little further, and the takeaway follows.

Figure 7.3 makes the contrast concrete.

The mathematics behind it: why copying multiplies routes *(this box can be skipped)*

When a node is duplicated, the copy and the original share all their neighbors, so any two of those neighbors sit at opposite corners of a four-cycle — each such pair instantly gains a second shortest route. A duplicated node of degree k creates $\binom{k}{2} = k(k-1)/2$ such squares in a single stroke. Moreover the duplication rule carries its own preferential attachment, the chance of copying a node being

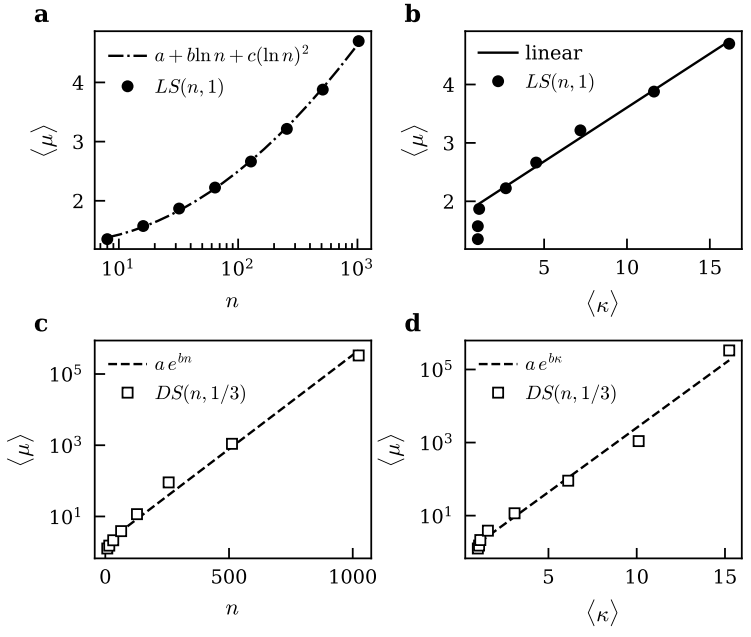
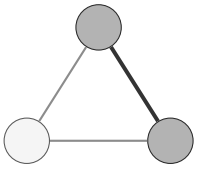
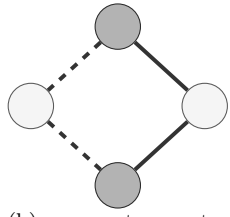


Figure 7.2: **Local rules boost the multiplicity.** (a) Local search $LS(n,1)$ grows log-quadratically, Eq. (7.29) — while its shuffled counterpart, with identical degrees, saturates (Figure 7.1a). (b) The same data against the inferred number of communities $\langle\kappa\rangle$: linear. (c) Duplication–split $DS(n,1/3)$ grows exponentially, Eq. (7.30); note the linear size axis and the logarithmic multiplicity axis, and that the vertical scale spans five decades. (d) Against $\langle\kappa\rangle$: exponential again. Communities are only detectable beyond the Ramsey size r_κ, so the leftmost points in (b) and (d) sit at $\langle\kappa\rangle = 1$.

proportional to its degree, which (as in Chapter 5) builds a heavy-tailed degree distribution with exponent $\gamma = 1/q$. Squares are thus laid down in proportion to $k(k-1)$ — the very combination that Eq. (7.1) says controls the multiplicity — on top of a distribution



(a) triangle: one route



(b) square: two routes

Figure 7.3: **Why squares multiply routes.** The two highlighted corners are the endpoints. (a) In a triangle they are joined directly — a single shortest route. (b) In a square they are opposite corners, two steps apart by *two* distinct routes (drawn solid and dashed), one around each side. Local rules that close four-cycles, such as gene duplication, thread the network with these redundant shortest paths.

with many high-degree nodes. Each round of copying compounds the squares laid down by the last, and it is this compounding, rather than any one square, that is the suspected origin of the exponential law, Eq. (7.30), though a full account of it remains open.

The lesson of the box is that squares, not triangles, are what multiply shortest routes. The bubble model lets us test that directly, since it dials the loop length by hand: a chain of L added nodes makes a loop of length $L + 2$. The geometric intuition survives the test (Figure 7.4a). Even-length loops, whose opposite corners enjoy two equal routes, enrich the network more than odd ones: $L = 2$ and $L = 4$ sit well above $L = 1$ and $L = 3$. But the test also sets a limit on how much the geometry explains. Every bubble model, whatever its loop length, stays on the log-quadratic rung. Even loops beat odd ones, and squares beat triangles — yet neither buys the exponential. The square-counting explains why duplication *outruns* triadic closure; the full ferocity of duplication–split needs the compounding above.

One more clue shows that randomness itself plays a role (Figure 7.4b). The Dorogovtsev–Goltsev–Mendes "pseudofractal" network (Dorogovtsev et al., 2002) is a *deterministic* relative of the

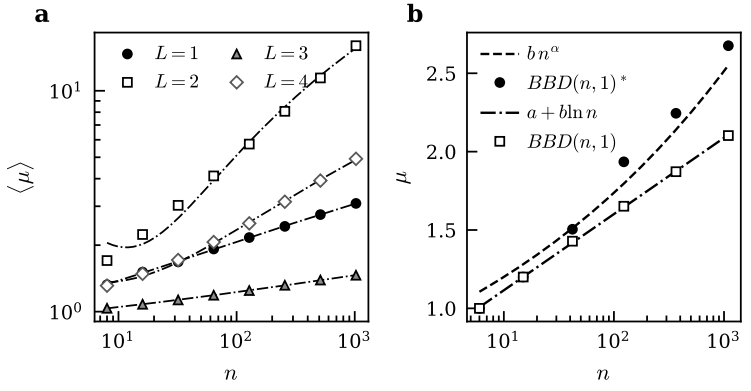


Figure 7.4: **Cycle length and randomness.** (a) The bubble model $BB(n, L)$, which closes loops of length $L + 2$. Even L (2 and 4) reaches a higher multiplicity than odd L (1 and 3), as the square-counting predicts — yet every L stays log-quadratic, so cycle geometry alone does not buy the exponential. (b) The deterministic bubble model $BBD(n, 1)$ grows only logarithmically, falling *below* its own randomized version $BBD(n, 1)^*$, which follows the power law of Eq. (7.20) with $\alpha \approx 0.16$. Locality without randomness is not enough.

bubble model, grown by a fixed clockwork rule rather than a random one: at every step, new nodes are added to *every* link at once. Its degree distribution is scale-free with $\gamma = 1 + \ln 3/\ln 2 \approx 2.6$, so by Eq. (7.20) its randomized version should show a power law with $\alpha \approx 0.16$ — and it does. The deterministic original, however, grows only logarithmically: it falls *below* its own shuffled counterpart. So the speed-up is not a property of the loop geometry alone; a clockwork rule that closes loops everywhere at once does worse than chance. It is the interplay of randomness and locality together, not either by itself, that boosts the multiplicity.

7.7 How fast could it possibly grow?

If randomness and locality together drive the multiplicity up, it is fair to ask where the ceiling is. There is one construction for which the question can be answered exactly, and it is a close relative of the bubble rule: the *diamond lattice* (Migdal, 1975; Kadanoff, 1976). In its general (b, s) form, start with a single link joining two poles A and B and, at each generation, replace *every* link by b parallel paths of s links each. The standard diamond is $b = s = 2$: every link becomes two parallel two-link paths (Figure 7.5a). Where the bubble model drops one bubble on one randomly chosen link, the diamond drops a bubble on all links at once, in perfect symmetry. It is the same lattice on which Chapter 6 watched the Ramsey community number become a renormalization-group crossing.

The mathematics behind it: the multiplicity between the poles *(this box can be skipped)*

The construction is exactly renormalizable, because distinct copies of the previous generation meet only at their corners. Read it backwards — deflation — and generation G_t is b parallel branches, each made of s copies of G_{t-1} in series. Two rules suffice. Along a series, lengths add and multiplicities *multiply*, since a route must make each choice in turn. Across b parallel branches of *equal* length, multiplicities *add*, since each branch offers its own routes and all tie. So the pole-to-pole distance D_t and multiplicity $P_t \equiv \mu_{AB}$ obey

$$D_t = s\, D_{t-1} = s^{\,t}, \qquad P_t = b\, P_{t-1}^{\,s}, \tag{7.31}$$

with $P_0 = 1$. The second is nonlinear, but taking $\log_b$ linearizes it, $\log_b P_t = 1 + s \log_b P_{t-1}$, a geometric recursion whose solution is

$$P_t = b^{\,(s^t-1)/(s-1)} \xrightarrow{\;b=s=2\;} 2^{\,2^t-1} = 2^{\,D_t-1}, \tag{7.32}$$

a *doubly* exponential count. (Equation (7.32) has been checked against direct construction of the lattice for (b, s) equal to $(2, 2)$, $(3, 2)$, $(2, 3)$, $(4, 2)$ and $(3, 3)$.) The reason is transparent: a shortest route must thread $(s^t - 1)/(s - 1)$ successive scales, and at each one it independently picks one of b parallel branches, so the choices multiply.

The poles are a special pair, so the average over *all* pairs needs its own argument — and it is here that the diamond's rigidity pays off again.

The mathematics behind it: from the poles to the average *(this box can be skipped)*

The key geometric fact is that every vertex lies on some pole-to-pole geodesic,

$$d(v, A) + d(v, B) = D_t, \tag{7.33}$$

proved by induction on t: within a replaced link the two poles of the sub-copy inherit the property, and any detour around a parallel branch is strictly longer. So a vertex is characterized by the single number $d(v, A)$ — the lattice is, in this sense, one-dimensional. Now split the pairs into those internal to one copy of G_{t-1} and those spanning two copies. The total geodesic count $S_t = \sum_{u<v} \mu_{uv}$ then satisfies

$$S_t = 4\, S_{t-1} + \mathrm{Inter}_t, \tag{7.34}$$

the four being the number of copies in the $b = s = 2$ cell, and Inter_t a convolution of the two copies' pole-path counts, weighted by the factor P_{t-1} that any route pays to cross the shared corner. Carrying the recursion out exactly gives the average $\langle \mu \rangle_t = S_t / \binom{n_t}{2}$ as

$$\langle \mu \rangle_t \;\sim\; \kappa\, \frac{2^{2^t}}{4^t} \;\approx\; 2.1\, \frac{P_t}{n_t}, \qquad \kappa \to 1.576, \tag{7.35}$$

where the prefactor κ is the fixed point of the corner-transit convolution. So the average merely tracks the pole value divided by the node count: the sum is dominated by near-antipodal pairs, which inherit a geodesic count of order $P_{t-1}^2 \sim P_t$.

Equation (7.35) is expressed in the generation t; the scaling with size follows by inverting the node count.

The mathematics behind it: the stretched exponential *(this box can be skipped)*

Generation t carries

$$n_t = \tfrac{1}{3}\left(2 \cdot 4^t + 4\right) \simeq \tfrac{2}{3}\, 4^t, \tag{7.36}$$

so $4^t \simeq \frac{3}{2}n$ and therefore $2^t \simeq \sqrt{3n/2}$. Substituting both into Eq. (7.35), the doubly exponential in t becomes a *stretched exponential* in n:

$$\langle\mu\rangle \approx a\, e^{\,b\sqrt{n}}, \qquad b = \sqrt{\tfrac{3}{2}}\ln 2 \approx 0.849, \qquad a = \tfrac{2\kappa}{3} \approx 1.05. \quad (7.37)$$

The rate b is exact; a is the amplitude of the leading term, since $\ln\langle\mu\rangle \simeq b\sqrt{n} + \ln a$ and the algebraic factor $1/n$ from Eq. (7.35) is subleading against $\sqrt{n}$. Retaining it, the reduced form $\langle\mu\rangle \simeq a\, e^{\,b\sqrt{n}}/n$ reproduces the exact values to better than 1% from $t = 4$ on.

The numbers are worth seeing, because they are exact rather than fitted (Figure 7.5b). By the sixth generation the diamond has only 2732 nodes, yet an average pair is joined by some 7×10^{15} distinct shortest paths. This is the fastest growth found anywhere in this chapter, and its origin is the same parallel branching that the local rules exploit — only here it is applied everywhere, at every scale, without a single random choice. Perfect, maximally symmetric branching therefore sets a hard ceiling on the redundancy a growing network can carry, and the stochastic local rules of the previous sections all fall below it.

7.8 Two sides of one coin

Because local rules also create communities, and the number of communities grows along with the network, we can ask how the multiplicity depends on the number of communities directly. The relationship is a clean one (Figure 7.2b and d): linear for every local model tested, and — once again the exception — exponential for duplication–split. This is the empirical correlation of Deng et al. (2026) restated from the inside: more communities go hand in hand with more shortest paths.

But the point of the chapter is that this partnership is not a case of one causing the other. “Local rules boost shortest path multiplicity” and “communities accompany high shortest path multiplicity” are two true statements about the same networks, and as descriptions they say the same thing. Only one of them, however, names a

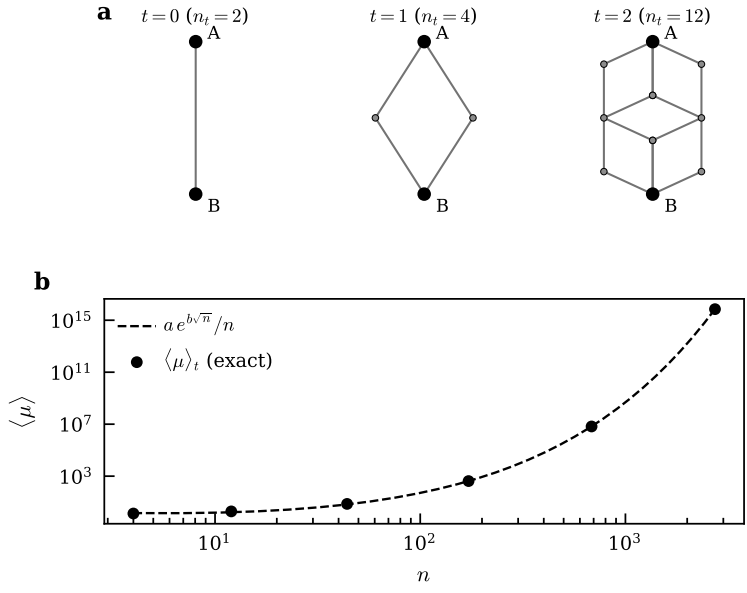


Figure 7.5: **The exactly solvable ceiling.** (a) The diamond lattice: every link is replaced, at every generation, by two parallel two-link paths. (b) Its average multiplicity, computed exactly, follows the stretched exponential of Eq. (7.37) across fifteen decades. With 2732 nodes, a typical pair of nodes is already joined by some 7×10^{15} equally short routes.

mechanism. Communities are something an algorithm reads off a finished network after the fact; the local rule is the process that actually built that network, node by node, closing the very loops that both carve it into communities and thread it with redundant routes. Community structure and path multiplicity are two sides of a single coin — and the coin is locality.

7.9 Real networks

Models are one thing; the networks we started from are another. Three systems can be caught at different stages of their growth, which is what we need if we are to watch $\langle\mu\rangle$ move with n (Figure 7.6). The Internet at the level of autonomous systems, mapped daily over several years (Leskovec and Krevl, 2014). The protein interaction networks of a dozen organisms (Stark et al., 2006), which stand in for successive stages of evolution. And the co-authorship network of the `cond-mat` preprint archive (arXiv.org submitters, 2024), accumulated year by year. In each case we can also compute the excess degree c_2 and so plot the mean-field expectation c_2/e alongside the measurement — the random network with the same degrees. The data are noisy and the systems are not stationary, so no strong claim is available; but the trends are instructive.

In every system the multiplicity rises with size, and in every system it is high: the human protein network reaches $\langle\mu\rangle \approx 64$, meaning a typical pair of proteins is joined by some sixty-four equally short chains of interaction. The protein networks are also the ones that climb *exponentially* with size, which is the signature of duplication — and gene duplication is, of course, exactly how they are thought to grow (Chapter 4). The Internet and the co-authorship network both climb as power laws, more sedately.

The most interesting feature, though, is which side of the mean-field line each system falls on, because the two disagree in a way that lines up with the degree correlations of Chapter 1. The Internet sits *below* its random expectation: it is *disassortative*, its hubs presiding over peripheries of poorly connected nodes, and such a star-like arrangement gives a route no alternatives — everything must pass through the hub. The co-authorship network sits *above* it: it is *assortative*, its well-connected authors writing with one another, and a clique of hubs is exactly what furnishes many equally short ways across. The same reading applies to the protein networks, which are disassortative and duly fall below the line. Degree correlations, in other words, modulate the redundancy on top of what the degrees alone would give — a suggestion the mean-field calculation, which knows only c_2, cannot yet account for. It is one of the clearer invitations this book leaves open.

7.10 Summary

The shortest paths of real networks are richly redundant, and that redundancy travels with their community structure. For a random network the redundancy is no mystery: it is the branching factor, $\langle\mu\rangle \approx c_2/e$, and it grows only as fast as the hubs allow — not at all when $\gamma > 3$, logarithmically at $\gamma = 3$, as a power law below it. Networks built by local rules leave that baseline behind, log-quadratically for most rules and exponentially for duplication, which closes the four-cycles that give opposite corners two shortest routes apiece. Neither cycle geometry nor locality is sufficient by itself: a deterministic rule that closes loops everywhere at once does worse than chance, while the perfectly symmetric diamond lattice, where the branching can be solved exactly, marks the ceiling at a stretched exponential. Since the same local rules also produce the communities, the observed tie between path multiplicity and communities is best understood not as cause and effect but as two consequences of one underlying local dynamics.

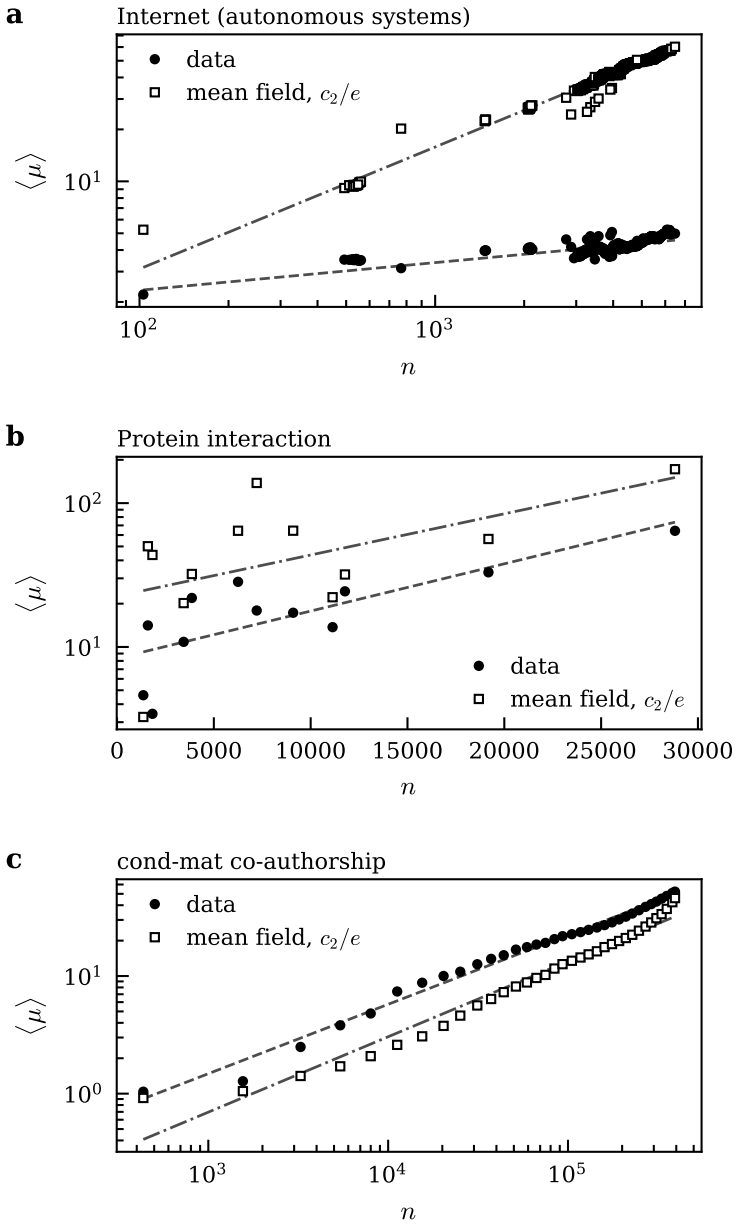


Figure 7.6: **Real networks.** Average multiplicity against size, with

Chapter 8

Outlook

8.1 One principle, many faces

We have now followed a single idea through six concrete settings, and it is worth gathering the threads before asking what lies beyond them. In every case the network grew by a *local* rule — a newcomer acting only on a node it had reached and that node's immediate neighbors — and in every case the same family of global regularities emerged unbidden.

Searchable networks, such as the citation graph and the World Wide Web, grow by exploration: a newcomer surfs along links and connects to what it finds. Modeling the surfer as the very random walk that defines Google's PageRank, we found that a page's visiting probability rises in a straight line with its in-degree, so that the local act of walking manufactures an effective "rich get richer." From it follow a power-law degree distribution whose exponent is fixed purely by how the network is explored, a clustering hierarchy in which cliquishness falls with degree, and generically disassortative correlations (Chapter 2).

Social networks grow by introduction: a friend introduces a friend, closing a triangle. Formalizing the friend-of-a-friend as a *potential link*, born in proportion to degree, we recovered the same rich-get-richer and the same clustering hierarchy — but now with *assortative* correlations, the well-connected keeping one another's

company (Chapter 3). The contrast with the searchable networks is that chapter's quiet lesson: the sign of the degree correlations is not a free parameter but a readout of whether the rule reaches outward, to the periphery, or inward, within a neighborhood.

Biological networks grow by copying: a gene is duplicated, its protein inheriting all of the ancestor's interactions, and the copies then diverge. Because a node gains a link whenever one of its neighbors is duplicated, copying too is a rich-get-richer engine; self-interactions seed the clustering hierarchy, and coupling divergence to duplication adds a density transition and a multifractal degree distribution born of inheritance, alongside the disassortative mixing measured in real protein networks (Chapter 4). Here the three mechanisms met in one sentence: a rule that attaches to a *neighbor* of a randomly chosen node attaches in proportion to degree, because a well-connected node simply belongs to more neighborhoods. Preferential attachment, the clustering hierarchy, and the degree correlations are all the generic harvest of locality.

The same copying, carried into project schedules, refines each activity either by duplicating it into parallel copies or by splitting it into specialized sequential parts. The rule predicts identical power laws for an activity's predecessors and successors, a critical path that vanishes as a fraction of large projects, and a fractal rather than small-world geometry — all borne out in seventy-seven real construction projects — and it relocates the risk of delay from the celebrated critical path to a percolation threshold set by the whole network's average degree (Chapter 5).

Finally, locality proved to shape structure above the level of the individual node. The same rules that set the degrees also segregate the network into communities — even when every node is identical — a fact we made precise through the Ramsey community number, the size beyond which communities are all but certain (Chapter 6); and they thread the network with redundant shortest paths, multiplying the routes between nodes far faster than chance and in lockstep with those very communities (Chapter 7).

And with that, the book comes full circle. We set out asking where the global regularities of real networks come from — their power-law degrees, their small-world distances, their clustering hierarchy, their degree correlations — and answered, one mechanism

at a time, that none of them need be imposed from above: each emerges from a way of growing in which no node ever uses more than local knowledge. A node that surfs to a neighbor, befriends a friend, is copied from an ancestor, or refines a task asks nothing of the network as a whole — and yet the networks these humble acts build carry, in the aggregate, the full and unmistakable signature of real complex systems.

8.2 What remains to be done

If the thesis of this book is right, it opens more questions than it closes.

Several are concrete. The exponential growth of shortest path multiplicity under the duplication–split rule — so much faster than the log-quadratic growth of every other rule we studied — still wants a full explanation; the square-counting argument shows why copying outruns triadic closure, but not why it runs away exponentially. The Ramsey community number, likewise, remains tied to the community-detection method used to measure it, and a notion of emergence that does not depend on the algorithm would put the whole idea on firmer ground. Even the self-organized critical state of the exhaustive searcher, where a scale-free network appears across a whole range of the control parameter rather than at one fine-tuned value, deserves to be connected more tightly to the wider physics of criticality.

Others are programmatic. Throughout, we read each local rule *forward*, from mechanism to statistics; the empirical chapters hint at the reverse problem — reading the rule *backward*, inferring from a single observed network which local moves grew it, and with what parameters, as we did when fitting a duplication index to project schedules. A general inverse theory, turning measured correlations, clustering hierarchies and community structure into an estimate of the underlying rule, would make these models genuinely diagnostic. And real networks are surely not grown by one rule in isolation: a social graph is part introduction and part search, a cellular network part duplication and part fresh interaction. How competing local moves combine — whether their signatures simply add, or interfere to produce something neither would alone — is largely open.

The setting, too, can be widened. The networks of this book are

mostly unweighted, undirected and static, whereas real ones carry weights, directions, timestamps, and several kinds of link at once. Whether the locality principle survives into weighted, temporal and multilayer networks — and what new regularities it might generate there — is a natural frontier. So is the coupling of growth to *function*: the percolation of delays through a project schedule was a first glimpse of a dynamics running on a network that local rules had built, and the interplay between how a network grows and what it then does promises more than either studied alone.

None of these questions threatens the central claim; each extends it. Local rules are the driving mechanism; everything else follows.

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